Is This X? A Null-Standardized Question Space for Corpus Curvature, with a Hodge-Theoretic Channel and a Calibrated Standard-Candle Generator

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Abstract

Thirty-four measurement cycles of a corpus-curvature program asked "is this X?" one X at a time — Ginzburg–Landau, LIGO, Grothendieck, pentagon — and answered PARTIAL every time, because no cycle computed what its statistic would have been under a matched null. This paper replaces the sequence of candidate X's with a map. Four corrections and three constructions are reported. (i) The headline observable $V_2 = PC1 + PC2$ is a decreasing function of feature-space dimension D at fixed rows, so the celebrated +0.4664 "phase transition" (0.2993 → 0.7657) is a basis change: on the identical 2286 rows we measure $V_2 = 0.7235$ at $D = 9$ and 0.2940 at $D = 24$. (ii) the shipped estimator reports not the participation ratio but the two-share proxy $\hat{r} = 2/V_2$, so the gate $V_2 \geq 0.40$ is *by definition* $\hat{r} \leq 5$ — an identity with no empirical content; what is empirical is that the exact-1.0000 gate passes at 2-D and at the 384-D $\left(\ell = 384, m = 3\right)$ variant are rank degeneracies reproduced with zero data content on all six corpora, including a single-class iid one. (iii) Under a fixed-margin (curveball) null drawn by two independent, provably uniform samplers (χ^2 uniformity $p = 0.364$ and $p = 0.506$ on exhaustively enumerated fibres), the 2286×9 matrix has null $V_2 = 0.7089 \pm 0.0014$ against a real 0.723529: $z = +12.3$. About 98% of V_2 is fixed by the marginals and the surviving signal is $\Delta V_2 = +0.0144$. An intermediate analysis reported 0.8005, $z = -45.8$; that value is retracted here after two independent replications. (iv) $\Delta V_2(N)$ is flat (+0.0135 to +0.0175) for $N \geq 79$ with no interior maximum, so the previous $N_c \approx 40$ was the null's own finite-size inflation and the transition does not exist. We add a Hodge channel (Jiang–Lim–Yao–Ye decomposition on the item graph): $\rho_g = 0.99961$, $\rho_c = 0.00013$, $\rho_h = 0.00026$, $\beta_1 = 6756$, and against a degree-matched null $z\left(\rho_h\right) = -53.1$, $z\left(\beta_1\right) = -312.5$ — the corpus is strongly *anti*-cyclic, which inverts the vortex reading; all three pre-registered Hodge predictions are falsified. The positive result is an instrument: a standard-candle generator planting a spherical-harmonic curve of amplitude s on a Fibonacci golden-angle lattice, on which the null-standardized statistic $\Delta V_2 z$ has detection power monotone in s over $s \geq 0.5$ at every grid point and a bounded false-positive rate ($s = 0$: $z \approx 0$, one-sided FPR 8.3% against a nominal 5%; $s = 1$, $N = 512$: $z = +3.43$, power 1.00; $s = 2$, $N = 2048$: $z = +28.07$), while at $s \leq 0.25$, and at $N = 128$ throughout, the cell means sit at the power floor and are not ordered while raw V_2 misorders corpora by signal. The deliverable is IS-THIS-X: a map Φ from corpus to null-standardized statistic vector, reference families, exclusion-only verdicts. yubiOS places as M_0-excluded ($z = +12.3$), nearest a two-population mixture with a small positive residual; mean-field Ginzburg–Landau is excluded. Two extensions close the paper. Aggregating a constant-ratio amplitude ladder — an *algebraic fold*, $s \in \{0.25, 0.5, 1, 2\}$ at $N = 128$, $d = 9$ — converts four cells that individually sit at the power floor into one monotone family test: the fold-slope statistic reaches $t = 45.19$ paired and $t = 28.92$ unpaired against a null-ladder 95% quantile of 0.0997, power 1.00 in both designs, which repairs clause (iii) at the family level without restoring per-cell monotonicity. And the map is run on the *real* GWTC catalogue (GWOSC, 269 events with all nine parameters, fetched 2026-08-12), which returns $V_2 = 0.8353$ against a column-shuffle null 0.2753 ± 0.0084, $z = +67.0$ — a z inflated by known astrophysical parameter redundancy — while the Y_3^3 Parseval energy share is 0.00334 against a null 0.00832 ± 0.00427, $z = -1.17$: not significant, an honest null result for the three-fold azimuthal lobe on real data. Five items the previous draft named as untested are closed here. Three close negatively — the 24-D curl clause ($\rho_s = +0.17$, $p = 0.69$), the three untested Ginzburg–Landau predictions (no fluctuation peak, no susceptibility divergence, no evidence of critical slowing-down), and the azimuthal channel, whose ordering-permutation null (500 index shuffles) leaves the Y_3^3 probe at $|z| \leq 0.19$ on both the 2286×9 master matrix and the real 269×9 GWTC matrix, with the $m = 3$ Parseval shares *below* their nulls because PC1 ordering starves high azimuthal order; a fourth — quantized harmonic winding — is void as posed, its winding half an identity of the wrapping convention; the fifth closes structurally: on the edit log, 1178 independently re-measured transitions contain zero regressions, detailed balance is violated at every level with non-zero flux, and the unique stationary distribution is the absorbing full-coverage state — so "emergence over cycles" is deterministic descent of a bounded exchangeable coverage potential to a fixpoint, with no equilibrium ensemble and hence no Ginzburg–Landau reading available in principle. A designed counterpart closes the loop: a quantized ± atom (one primitive flip per move) with Metropolis–Hastings acceptance on the same measured Φ ladder restores an exact equilibrium ensemble $\pi_T(k) \propto \binom{9}{k} e^{-\Phi\left(k\right)/T}$ with free energy $F_T(k) = \Phi(k) - T \log \binom{9}{k}$ — detailed balance not rejected at $\max_k |z\left(J\right)| = 0.010$ over 800,000 samples — whose $T \to 0$ limit reproduces the historical absorbing log (zero backward transitions, mass 1.00000 at $k = 9$), locating the free-energy language as a property of designable dynamics rather than of the corpus.

Introduction

The predecessor of this paper (Tchatalbachian 2026a) fitted learned latent curves to a binary corpus-coverage matrix and reported one fit-quality gate: the top-2 eigenvalue share $V_2 = PC1 + PC2 \geq 0.40$. Thirty-four subsequent measurement cycles took that number as an order parameter and asked, in sequence, whether the corpus *is* a Ginzburg–Landau system (Tchatalbachian 2026b; Hohenberg & Krekhov 2015), a gravitational-wave catalogue, a Grothendieck-gap instance, a pentagon. Each cycle answered by narrative compatibility. The median verdict was PARTIAL.

The defect is structural. A question space with candidate classes but no origin cannot weigh "compatible with X" against "compatible with nothing". Every class was tested with bespoke estimators — saturation windows, kernel bandwidths, row-lag correlation lengths — not comparable across X, across corpora, or even across cycles; the same κ estimator returned 1.06 on one corpus and 323 on another, the latter a saturation-clamp artifact. The fix is therefore not another X. It is a map,

$$\Phi\text{: corpus} \mapsto s \in \mathbb{R}^m, \qquad s \text{ null-standardized}, \qquad (1)$$

with the matched-marginal null at the origin, invariance requirements that delete non-invariant estimators by construction, and exclusion-only verdicts. Equation (1) is the thesis; Sections 3–6 are its content.

Section 2 restates the predecessor's governing equations, with the Y_3^3 normalization corrected. Section 3 gives the null models and the corrected record, including a retraction. Section 4 gives the Hodge channel and its three falsified predictions. Section 5 gives the standard-candle generator and calibrated detection — the paper's positive result. Section 6 defines the IS-THIS-X question space and places yubiOS and a GWTC-like corpus in it. Sections 8–9 are scope and conclusion; Appendix A is the reproduction manifest.

Two disciplines run throughout. No statistic is reported without the value it takes on a matched null. And when a number in the internal record fails to reproduce, the failure is printed rather than the number quietly replaced — Section 3.3 is that discipline applied to this program's own most-cited intermediate result.

Background: the learned-latent-curve family

Flat curve, ridge solve, gate

For output dimension $j = 1, \ldots, D$ and a 1-D ordering coordinate $t \in \left[0, 1\right]$ (Tchatalbachian 2026a),

$$z_j(t) = a_{j,0} + \sum_{m=1}^{k} \left(a_{j,m} \sin(2\pi f_m t) + b_{j,m} \cos(2\pi f_m t)\right), \qquad (2)$$

with k shared learned frequencies. Stacking the design vector

$$\phi(t) = [1,\ \sin(2\pi f_1 t),\ \cos(2\pi f_1 t),\ \ldots,\ \sin\left(2\pi f_k t\right),\ \cos\left(2\pi f_k t\right)] \in \mathbb{R}^{1+2k} \qquad (3)$$

gives $\gamma(t) = C\phi(t)$, solved at fixed frequencies by the Tikhonov ridge

$$C^\star = \left(\Phi^\top \Phi + \lambda I\right)^{-1} \Phi^\top Z, \qquad \lambda = 10^{-3}, \qquad (4)$$

still the sanity floor for any gradient fit at the same frequencies. The coordinate t is the rank of the top principal component of the z-scored feature matrix, and the gate is

$$V_2 \; = \; PC1 + PC2 \; = \; \frac{\lambda_1 + \lambda_2}{\sum_{a=1}^{D} \lambda_a} \; \geq \; 0.40, \qquad (5)$$

the λ_a being eigenvalues of the $D \times D$ correlation matrix, so $\sum_a \lambda_a = D$. Equation (5) is the observable this paper audits.

Hyperspherical variant and Möbius chart

For $\mathbf{x} \in S^2$,

$$\gamma(\mathbf{x}) = \mathbf{b} + \sum_{\ell=0}^{L} \sum_m \mathbf{a}_{\ell,m} Y_{\ell,m}^{S^2}(\varphi_\theta(\mathbf{x})), \qquad n_{basis} = \sum_{\ell=0}^{L} \left(2\ell + 1\right) = 16 \text{ at } L = 3, \qquad (6)$$

with the Möbius reparameterization

$$\varphi_\theta(z) = \frac{az + b}{cz + d}, \qquad a, b, c, d \in \mathbb{C}, \qquad ad - bc \; = \; +1, \qquad (7)$$

carrying six real degrees of freedom (eight real components less the two real constraints of one complex equation; the $M \sim -M$ identification of $PSL\left(2, \mathbb{C}\right)$ is discrete and removes none). In every run reported here and in the predecessor's appendices φ_θ is *frozen at the identity*; no L-BFGS-B refinement is performed. Items reach S^2 by the stereographic lift of their PCA top-2 coordinates,

$$\sigma\left(u, v\right) = \frac{1}{1 + u^2 + v^2}(2u,\ 2v,\ u^2 + v^2 - 1) \in S^2. \qquad (8)$$

Fibonacci sampling and the corrected Y_3^3 constant

Sampling on S^2 uses Vogel's golden-angle lattice (Vogel 1979; Saff & Kuijlaars 1997):

$$z_i = 1 - \frac{2i+1}{N}, \qquad \phi_i = 2\pi \frac{i}{\varphi_g}, \qquad \theta_i = \arccos(z_i), \qquad \varphi_g = \frac{1+\sqrt{5}}{2}, \qquad (9)$$

so the corpus item at index i *is* the parameter point ($i = t$, no lookup table, $O(1)$ per point, no pole clustering). The angular probe is the three-fold azimuthal harmonic

$$Y_3^3\left(\theta, \phi\right) \; = \; K \; \sin^3 \theta \; \cos\left(3\phi\right). \qquad (10)$$

The constant is corrected here. With Condon–Shortley phase (NIST DLMF),

$$K_{\mathbb{C}} = \sqrt{\frac{35}{64\pi}} = 0.417224, \qquad K_{\mathbb{R}} = \sqrt{\frac{70}{64\pi}} = 0.590044, \qquad (11)$$

$K_{\mathbb{R}}$ being the real-orthonormal form appropriate to a real basis. The legacy regime constant $K_{legacy} = \sqrt{245/(64\pi)} = 1.103870$ is wrong: since $245/35 = 7$ and $245/70 = 3.5$,

$$\frac{K_{legacy}}{K_{\mathbb{C}}} = \sqrt{7} = 2.645751, \qquad \frac{K_{legacy}}{K_{\mathbb{R}}} = \sqrt{7/2} = 1.870829. \qquad (12)$$

The error is the factor $\left(2\ell + 1\right) = 7$ counted twice — it is already inside $35/64\pi$. Every Y_3^3 magnitude published under the legacy constant is inflated by $1.870829\times$. The correction is scale-only: because the generator of Section 5 standardizes each probe channel before scaling by the planted amplitude, switching constants leaves the emitted matrix bit-identical and V_2 identical to 10^{-12} (machine-asserted, generator self-test step 4).

Remark 1 (what φ_g and Y_3^3 cannot claim). The Fibonacci lattice is a sampling scheme and Y_3^3 a fixed basis element used as a probe; neither is a model of the corpus. Two claims are inadmissible without further work. *Azimuthal symmetry*: ϕ_i is assigned by construction from the index and the index is the PC1 rank, so power at $\cos\left(3\phi\right)$ is a property of the index map until an ordering-permutation null says otherwise. That null has now been executed (500 shuffles of which row receives which lattice point, matrix and lattice untouched), and it says otherwise in the direction opposite to the claim: see Section 8. *The 384-D gate pass*: $PC1 + PC2 = 1.0000$ on the $\left(\ell = 384, m = 3\right)$ variant is a rank artifact (Proposition 2), not a fit. The golden ratio in the sampler licenses no numerology: it has no relation to any fitted κ, to pentagon geometry, or to 7/9.

Null models and the corrected record

Throughout, $M \in \{0,1\}^{N \times D}$ is the coverage matrix and $c_i = \sum_a M_{ia}$ the row mass. The primary null is the *fixed-margin* (curveball) ensemble (Strona et al. 2014; Carstens 2015): the uniform distribution on the fibre of binary matrices with the row sums *and* column sums of M. The primary statistic is the null-standardized residual

$$\Delta V_2 z \; = \; \frac{V_2 - \mathbb{E}_0[V_2]}{SD_0[V_2]}, \qquad \Delta V_2 = V_2 - \mathbb{E}_0[V_2], \qquad (13)$$

with $\mathbb{E}_0, SD_0$ over the curveball ensemble at the corpus's own N, D and marginals.

C1 — V_2 is a dimension artifact

Proposition 1 (trace-normalized share decays in D). *Let the rows be fixed and let $M^{(D)}$ append $D - D_0$ further columns to D_0 original ones. On the correlation matrix* $\operatorname{tr}\Sigma^{(D)} = D$ *for every D, so $V_2^{(D)} = (\lambda_1 + \lambda_2)/D$. An appended column contributes unit trace and, unless it loads on the existing leading plane, contributes $O(1)$ to $\sum_a \lambda_a$ and $o(1)$ to $\lambda_1 + \lambda_2$. Hence $V_2^{(D)}$ is non-increasing in D up to the leading-plane loading of the new columns, with flat-spectrum floor $V_2 \to 2/D$.*

Proof sketch. z-scoring forces $\Sigma_{aa} = 1$, so $\operatorname{tr}\Sigma = D$ and V_2 is an eigenvalue *share*, not an eigenvalue. In the limit of exact independence, adding one column adds an eigenvalue 1 to the denominator and nothing to the numerator, giving $V_2 \mapsto V_2 \cdot D_0/(D_0 + 1)$; correlated additions interpolate between that dilution and no change. For an exactly flat spectrum $\lambda_a \equiv 1$, $V_2 = 2/D$, which is therefore the iid floor. □

Measured consequence, on the *identical* 2286 rows (Table 1): $V_2 = 0.7235$ at $D = 9$ and $V_2 = 0.2940$ at $D = 24$ with independent nuisance columns (0.4473 when the nuisance columns are made row-mass-correlated). The cited transition of +0.4664 from 0.2993 to 0.7657 spans the same interval, with the 0.2993 baseline the 24-D basis and the endpoint the 9-D basis. There is no transition to explain: it is Proposition 1 evaluated at two points.

basis	D	$2/D$	yubiOS	2-class	1-class (iid)	3-class	planted	GWTC-like
PCA top-2	2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
sphere lift	3	0.6667	0.8302	0.7187	0.6843	0.6885	0.6865	0.8460
7-column subset	7	0.2857	0.7575	0.6573	0.3028	0.5923	0.3305	0.7346
9-D native	9	0.2222	0.7235	0.6454	0.2397	0.5575	0.2705	0.6624
16 real SH ($L = 3$)	16	0.1250	0.4857	0.4244	0.2202	0.3490	0.2227	0.4180
24-D indep. nuisance	24	0.0833	0.2940	0.2718	0.0984	0.2341	0.1064	0.2523
24-D mass-corr. nuis.	24	0.0833	0.4473	0.4221	0.1410	0.3620	0.1457	0.2852
384-D native (indep.)	384	0.0052	0.3828	0.2657	0.0370	0.2220	0.0358	0.0228
384-D native (replicated)	384	0.0052	0.3921	0.2567	0.0496	0.2391	0.0462	0.0451
384-D variant 1 ($\ell = 128$)	384	0.0052	0.7592	0.1569	0.0763	0.2241	0.1012	0.0594
384-D variant 2 ($\ell = 384, m = 3$)	384	0.0052	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**

Table 1. V_2 against feature-space dimension; rows held fixed within each column. Eleven bases; the 2-D and $\left(\ell = 384, m = 3\right)$ bases return exactly 1.0000 on every corpus, including the single-class iid one — zero data content. The $2/D$ column is the flat-spectrum floor of Proposition 1, derived for an exactly flat spectrum only; it is not a floor for an arbitrary corpus, and several entries sit either side of it. Effective ranks quoted for this table are the proxy $\hat{r} = 2/V_2$ of Eq. (15). Source: results/ladder_VD.json.

C2 — the gate is a rank test

Proposition 2 (the gate is a rank identity under the shipped estimator). *Let*

$$PR = \frac{\left(\sum_a \lambda_a\right)^2}{\sum_a \lambda_a^2} \qquad (14)$$

be the participation ratio — the quantity ordinarily meant by "effective rank", and the one computed by the generator's placement code (skills/curved-corpus-create/scripts/create_corpus.py). For a spectrum flat on its top-r block and negligible elsewhere, $PR = r$ and $\lambda_1 + \lambda_2 = (2/r)\sum_a \lambda_a$, so in that regime $V_2 \approx 2/PR$. The analysis code shipped with this paper (scripts/common.py) does not compute PR. It reports the two-share proxy

$$\hat{r} = \frac{2}{V_2}, \qquad (15)$$

a deterministic, strictly decreasing transform of V_2 and nothing more. Under that estimator the gate equivalence

$$V_2 \geq 0.40 \Leftrightarrow \hat{r} \leq 5 \qquad (16)$$

is a definitional identity: it is true by algebra, it carries no empirical content, and $\hat{r}$ adds no information to V_2. In particular $V_2 = 1.0000 \Leftrightarrow \hat{r} = 2$, which holds exactly when the sampled design has numerical rank 2 — attainable with no data content whatsoever.

Because Eq. (16) is an identity, none of the empirical weight sits there. What is empirical is *where* the identity bites, and that is the last row of Table 1: the $\left(\ell = 384, m = 3, \sin^{384}\theta\right)$ variant returns exactly 1.0000 — hence $\hat{r} = 2.00$ — on all six corpora of that table: real yubiOS, one-, two- and three-class latent mixtures, a planted-curve corpus, a GWTC-like corpus. Counting the second planted variant carried in the data file but not tabulated, results/ladder_VD.json holds 14 rows at $V_2 = 1.0000$ to machine precision, every one of them in the 2-D PCA basis or in the $\left(\ell = 384, m = 3\right)$ variant, and none anywhere else. A single-class iid corpus — pure noise, no latent structure at all — passes the gate at exactly 1.0000 in two of the eleven bases tried. That is a fact about the bases, not about the corpora, and it is what Proposition 2 is for. Three hundred and eighty-four deterministic functions of one scalar index span a two-dimensional column space by construction; the 2-D PCA row says the same thing trivially. The predecessor's 7-D $PC1 + PC2 = 1.0000$ on refs/ is the same artifact at the other end, its surviving coverage matrix being rank 2. Proxy values along the yubiOS ladder are $\hat{r} = 2.00$ ($D = 2$), 2.41 (3), 2.64 (7), 2.76 (9), 4.12 (16), 6.80 (24); by Eq. (15) each is $2/V_2$ and restates the yubiOS column of Table 1 rather than adding to it. A gate value of exactly 1.0000 is a red flag, not a success: the numerical rank and condition number of the design must be reported beside it, and the accept/reject decision made on $\Delta V_2 z$, which is dimension-comparable and marginal-matched.

Which effective rank is which. Two distinct quantities appear in this literature under the name "effective rank", and every table here states which one it uses. The values in Table 1 and the 3.02 / 8.02 quoted for the GWTC-like corpus in Section 5 are the proxy $\hat{r} = 2/V_2$ of Eq. (15), computed by scripts/common.py (verified: $\hat{r} = 2/V_2$ to $\sim 10^{-15}$ on all 77 rows of ladder_VD.json). The coordinate PR in the statistic vector of Eq. (42), and the $PR = 6.837$ quoted for the worked planted corpus in Sections 5 and 6, are the participation ratio of Eq. (14), computed by create_corpus.py. The two disagree, and must: on that corpus $V_2 = 0.4477$, so $\hat{r} = 4.47$ while $PR = 6.837$. Only PR is an independent coordinate of the map; $\hat{r}$ is not, and it survives in Table 1 as a readability aid only.

C3 — the corrected null, and a retraction

quantity	value	z (real vs null)	source run
real V_2 (2286×9)	0.7235293730732693	—	A2-curveball-audit
published cycle-20 value	0.723529 (reproduces exactly)	—	A2-curveball-audit
curveball null, $200 \times 228,600$ trades	0.709180 ± 0.001183	+12.13	A2-curveball-audit
independent swap-chain sampler	0.708593 ± 0.000507	+29.44	validation
row-marginal-only null	0.670823 ± 0.001073	+49.14	A2-curveball-audit
column-permutation null	0.239653 ± 0.002253	+214.79	A2-curveball-audit
iid Bernoulli (full entry shuffle)	0.239832 ± 0.002340	+206.70	A2-curveball-audit
row-permutation control	$0.723529 \pm 2.1 \times 10^{-16}$ (identity check)	n/a	A-v2-nulls
Marchenko–Pastur baseline ($q = 0.003937$)	$\lambda_{\pm} = 0.878446/1.129428$; top-2 share 0.241375	—	A-v2-nulls

Table 2. Nulls for the 2286×9 coverage matrix. Each row is a single run, named in the last column (files under results/), and each z is recomputed from the mean and sd printed in that same row against the real $V_2 = 0.7235293730732693$; an earlier version of this table spliced mean, sd and z from different runs. The two fixed-margin samplers agree on the mean (0.709180 against 0.708593, a gap of 0.0006) and disagree on the spread: the swap chain's sd is $2.3\times$ tighter, and that difference in sd is the entire difference between $z = +29.44$ and $z = +12.13$. The headline figure quoted throughout this paper is the curveball value +12.13, the more conservative of the two. The row-permutation row is an identity check — V_2 is invariant under row relabeling — not a null, and no z is defined for it.

Two independent samplers, each proved uniform on the exact row-and-column-marginal fibre, agree. Sampler A: on a 5×4 matrix whose fibre has exactly 120 elements, 300,000 draws hit all 120, hit nothing outside, and are uniform ($\chi^2 = 123.758$, $df = 119$, $p = 0.364$). Sampler B: on the exhaustively enumerated 6×4 ensemble (120 matrices, 20,000 draws), $\chi^2 = 118.096$, $df = 119$, $p = 0.506$. Marginals were asserted exactly on every one of the 2286×9 draws. Mixing is converged: V_2^{null} against trade count runs 0.7159258 ($1N$), 0.7103934 ($5N$), 0.7087008 ($20N$), 0.7092413 ($100N$), 0.7094042 ($400N$), at 12 replicates per point — flat from $20N$ on, within a null sd of ≈ 0.0012. These five values are the mixing_convergence block of results/A2-curveball-audit.json verbatim; the figures printed in an earlier draft (0.715314, 0.710096, 0.708886, 0.708878, 0.709364) appear in no shipped artifact and are withdrawn — the same defect this section retracts, and it is corrected here rather than left standing. The Marchenko–Pastur prediction 0.241375 (Marchenko & Pastur 1967) lands on top of the two destroyed-dependence nulls (0.2397, 0.2398), an independent check that the pipeline is sound.

The conclusion is quantitative and modest:

$$\Delta V_2 = 0.723529 - 0.709180 = +0.014415, \qquad \Delta V_2 z = +12.13, \qquad \frac{\mathbb{E}_0[V_2]}{V_2} = 0.9802. \qquad (17)$$

About 98% of the headline observable is fixed by row and column marginals alone; the surviving corpus-specific signal is +0.0144 of eigenvalue share. It is real ($z = +12$, not $|z| < 3$), and it is small.

Remark 2 (retraction). An intermediate internal analysis reported the same curveball null as 0.8005 ± 0.0017, hence $z = -45.8$ — the corpus *below* its own null, "anti-structured". That value does not reproduce and its sign is wrong. Two independent samplers, both validated as above and both converged in mixing, return 0.7089; 0.8005 is not reachable by longer mixing. Diagnostically, the same memo's other nulls do reproduce with the present code (column-permutation claimed 0.2410, measured 0.239733 ± 0.002; row-mass-only claimed 0.6709, measured 0.670800 ± 0.001), which localizes the error to its curveball implementation; the memo is additionally self-inconsistent, reporting 0.7613 for the same null at the same N in a second table. Every downstream statement resting on $z = -45.8$ — in particular "the corpus is less concentrated than chance" — is withdrawn. This is the paper's null discipline operating on its own lineage: the retraction is reported, not repaired silently.

Per-corpus residuals under the same null (50 samples each) are in Table 3. The smallest corpus, docs ($N = 126$), is indistinguishable from its own null at $z = -0.10$; the all-zero refused regime (204 identically zero rows, every column of zero variance) has no defined V_2 and is reported as a skip, not as 0.0.

corpus	N	V_2	curveball null	z	col-perm null	MP pred.
yubiOS whole	2286	0.723529	0.7098 ± 0.0011	+12.12	0.2399	0.2414
yubiOS eligible ($c \ge 5$)	1322	0.425567	0.3824 ± 0.0051	+8.41	0.2461	0.2475
yubiOS deferred ($1 \le c \le 4$)	760	0.311625	0.2596 ± 0.0045	+11.65	0.2527	0.2558
yubiOS refused ($c = 0$)	204	undefined	—	—	—	—
yubiOS skills	474	0.581825	0.5559 ± 0.0122	+2.12	0.2619	0.2650
yubiOS docs	126	0.607322	0.6080 ± 0.0064	−0.10	0.2965	0.3073
yubiOS refs	791	0.648135	0.5801 ± 0.0059	+11.51	0.2532	0.2551
yubiOS self	895	0.406537	0.3863 ± 0.0035	+5.81	0.2506	0.2531
yubiOS 1391 (cycle-18)	1391	0.640256	0.5882 ± 0.0046	+11.28	0.2456	0.2469
CIS 447 (dev-sec InSpec)	447	0.331961	0.2667 ± 0.0044	+14.93	0.2635	0.2663
SCAP SSG 2107	2107	0.384629	0.3422 ± 0.0020	+21.54	0.2411	0.2422
SCAP SSG rules-dir 700	700	0.368366	0.3274 ± 0.0032	+12.92	0.2550	0.2572

Table 3. V_2 against the fixed-margin null, per corpus. Source: results/A-v2-nulls.json.

C4 — there is no critical point

The previous program located $N_c \approx 40$ as the argmax of $dV_2/d\log N$ on *raw* V_2. Table 4 runs the same sweep with the null computed at each N (matched marginals, matched N, 8 subsamples per N, 120–200 null reps per point).

N	V_2 **real**	V_2 **null (curveball)**	ΔV_2	z	**col-perm null**
40	0.7501	0.7443	+0.0058	+0.6	0.3674
79	0.7459	0.7284	+0.0175	+2.5	0.3237
160	0.7321	0.7146	+0.0175	+3.7	0.2922
320	0.7270	0.7134	+0.0135	+4.3	0.2706
640	0.7197	0.7045	+0.0153	+6.5	0.2564
1280	0.7214	0.7069	+0.0145	+9.2	0.2462
2286	0.7235	0.7092	+0.0144	+12.3	0.2401

Table 4. Finite-size sweep with the null's own N-dependence exhibited. ΔV_2 is flat for $N \geq 79$; the growth of z is the shrinking null SD, not a growing effect. Sources: results/finite_size.json, results/finite_size_colperm.json, results/scaling_VN.json.

Two fitted laws separate the nulls:

$$V_2^{curveball}(N) = \frac{2}{D} + 0.5252\left(\frac{D}{N}\right)^{0.0167}, \qquad V_2^{colperm}(N) = \frac{2}{D} + 0.3117\left(\frac{D}{N}\right)^{0.5177}, \quad R^2_{colperm} = 0.99984. \tag{18}$$

The R^2 belongs to the column-permutation fit alone: finite_size.json reports $c = 0.5252344$ and $\alpha = 0.0167256$ for the curveball law but no R^2, and that law under-predicts the measured null by 0.008–0.010 at both ends of the sweep ($N = 40$: 0.7345 fitted against 0.7443 measured; $N = 2286$: 0.7010 against 0.7092), so it is a shape summary, not a calibrated fit. The column-permutation (iid) null relaxes toward $2/D$ like Marchenko–Pastur in the aspect ratio D/N; the fixed-margin null relaxes toward the same asymptote only logarithmically slowly — $\alpha = 0.0167$ makes the approach invisible over the sampled range, because row masses are held — so for practical purposes it does not relax at all here. So the "V_2 inflation at small N" story applies to the iid ensemble only. Window slopes make the point directly: the largest $|d(\text{raw } V_2)/d\log N|$ is -0.0196 in the 79–160 window, the largest null slope is -0.0234 at 40–79, and the largest null-subtracted slope is $+0.0172$, also at 40–79 and *positive*. $\Delta V_2(N)$ has no interior maximum; the previously reported $N_c \approx 40$ was raw V_2 inheriting the null's finite-size inflation at the smallest N in the sweep.

Generated corpora with known ground truth confirm that this flatness is the estimator working, not a blind spot. A one-class (iid) generator gives $|z| \leq 1.5$ at every N from 40 to 4096. A two-class mixture gives $\Delta V_2 = +0.0108, -0.0004, +0.0064, +0.0041, +0.0017, +0.0029, +0.0042, +0.0049$ across $N = 40, \ldots, 4096$ — a constant small offset whose z climbs from +0.9 to +7.7 purely through the shrinking null SD, which is exactly the yubiOS pattern.

GL prediction (internal entry, §4)	**pre-pivot status**	**post-null verdict**	**basis**
P1 saturation $V_\infty \approx 0.77$–0.79	survived	**reinterpreted**	V_∞ is the null value, not a constant
P2 critical point at max $dV_2/d\log N$	claimed	**falsified**	$\Delta V_2(N)$ flat, no interior max
P3 exponent $\beta_{emp} \approx 1/2$	not fitted	**vacated**	no transition to fit
P4 fluctuation peak $Var[V_2]/V_2$	not run	**falsified**	no interior peak once $Var[\Delta V_2]$ is divided by the null's own variance
P5 susceptibility divergence $\chi(N)$	not run	**falsified**	smooth power law, no divergence; the $1/N$ expectation also rejected
P6 type-I/II via $\kappa = \lambda/\xi$	survived	**reinterpreted**	κ estimators are row-order dependent
P7 critical slowing-down	not run	**not supported**	no evidence for a critical exponent; ν's CI includes 1 at both N
P8 universality collapse across corpora	claimed	**not supported**	per-corpus z spans −0.10 to +21.5

Table 5. Ginzburg–Landau scoreboard. Two of eight predictions were recorded as surviving before the pivot; both are yubiOS-specific and both are reinterpreted here. The κ family is deleted by the invariance requirement of Section 6: a row-lag estimator changes under row relabeling, and the same estimator returned $\kappa \approx 1.06$ on yubiOS and $\kappa = 323$ on NIST, the latter a saturation-clamp artifact. $V_{floor} = 0.222$ is 2/9 plus finite size; $V_\infty \approx 0.769$ is an ensemble value.

P4 (fluctuation peak). $Var[\Delta V_2](N)$ is measured over 40 row-subsamples per N on the grid $N = 40, 79, 160, 320, 640, 1280$, each subsample standardized against its own 20-rep curveball null. The raw variance is reported here only as a ratio, because a variance computed by subsampling a single finite corpus carries a built-in decay and has no calibrated baseline: the quantity in the second column below is $Var[\Delta V_2](N)$ divided by the variance of V_2 across the curveball draws *at the same* N, and the finite-population factor $(1/N - 1/2286)$ is printed beside it.

N	$Var[\Delta V_2]$	**null** $Var[V_2]$	**ratio**	**ratio CI**$_{95}$
40	1.394×10^{-4}	1.162×10^{-4}	1.20	$[0.69, 1.71]$
79	1.422×10^{-4}	4.719×10^{-5}	3.01	$[1.81, 4.13]$
160	5.679×10^{-5}	2.166×10^{-5}	2.62	$[1.52, 3.79]$
320	3.189×10^{-5}	9.971×10^{-6}	3.20	$[1.93, 4.61]$
640	1.516×10^{-5}	5.122×10^{-6}	2.96	$[1.68, 4.31]$
1280	8.215×10^{-6}	2.444×10^{-6}	3.36	$[2.26, 4.33]$

Table 6. The P4 fluctuation sweep, normalized. The third column is the curveball null's own variance in V_2 at the same N, averaged over the 40 reps; the ratio is the second column divided by it. The finite-population baseline $(1/N - 1/2286)$ falls by $71\times$ across this grid, the measured variance by $17\times$ and the null's variance by $48\times$, so the analytic baseline is not the right calibration and the measured null variance is. $N = 2286$ is off the grid: subsampling 2286 of 2286 rows is deterministic, so its residual spread is null-estimator noise and cannot enter a monotonicity claim. Source: results/p4-variance-ratio.json, regenerated by scripts/p4_variance_ratio.py.

The normalized fluctuation has no interior peak. Its argmax sits at the upper edge of the grid ($N = 1280$), the 40–79 band carries the *smallest* normalized fluctuation on the grid (mean ratio 2.11 against 3.04 over $N \geq 160$), and from $N = 79$ upward the ratio is flat at ≈ 3 within overlapping bootstrap intervals. The flat plateau in the raw variance over $N = 40$–79 — the one feature that would have looked like a fluctuation peak sitting on the retired N_c band — is the null's own variance, which is $2.5\times$ larger at $N = 40$ than at $N = 79$; dividing it out removes the plateau rather than sharpening it into a peak. P4 is falsified: there is no fluctuation peak beyond the finite-population subsampling baseline, and none at the retired $N_c \approx 40$–79 either. Two open readings are conceded: the corpus fluctuates at roughly $3\times$ its null's variance uniformly for $N \geq 79$, which is unexplained here, and the ratio's argmax at the grid edge ($N = 1280$) means absence of a peak beyond the grid is not established.

P5 (susceptibility). Single-row leave-one-out response, up to 200 rows per N over $N = 40$–2286, each LOO set against its own curveball null, gives a smooth monotone power law with no interior peak and no blow-up at any finite N: per-row response $= 0.0386\,N^{-0.585}$, 95% CI $\left[-0.648, -0.521\right]$, $R^2 = 0.991$. The two conventions for χ are the same fit in different units — with $h = 1/N$, $\chi = N \times$ (per-row response) $= 0.0386\,N^{+0.415}$, the exponents differing by exactly one — so neither the growth under one nor the decay under the other is evidence about criticality; what P5 requires, a divergence at a finite N_c, is absent in both. Two further remarks are owed. The naive expectation carried through this lineage, that one row's influence dilutes as $1/N$, is also rejected: -1 lies far outside the CI. And $-0.585 \approx -1/2$ is the scaling of a sampling fluctuation, so the LOO response at these N is plausibly dominated by estimator noise rather than by systematic response — which would mean P5 as posed had no discriminating power to begin with. Separating the two requires the same LOO sweep on matched null matrices, and that is not run here.

P7 (critical slowing-down). Recovery of ΔV_2 after removing 5% of rows and restoring a fraction f of them is fitted with a linear (append-only saturation) model and a critical-exponent model at $N = 2286$ and $N = 320$. The critical-exponent model buys $\leq 1\%$ of RSS and is not distinguishable from linear ($\Delta AIC = -2.00$ and -0.97); ν's 95% CI includes 1 at both N ($\left[-0.20, 2.23\right]$ and $\left[-0.01, 1.17\right]$). P7 is not supported, and no positive claim for the linear form is made either.

The Hodge channel

The second measurement channel is combinatorial Hodge theory (Jiang et al. 2011), adopted because it supplies what an eigenvalue share cannot: a decomposition with an exact orthogonality guarantee, independent of PCA, of the stereographic lift, of λ, and of the Möbius chart.

Operators

Let $V = \{1, \ldots, N\}$ be items and Λ the voters (primitives). Each voter contributes a skew-symmetric matrix and an indicator weight, aggregated as

$$Y_{ij}^{\alpha} = -Y_{ji}^{\alpha}, \quad w_{ij}^{\alpha} = 1\left[\alpha \text{ compared } \{i, j\}\right], \qquad w_{ij} = \sum_{\alpha} w_{ij}^{\alpha}, \quad \bar{Y}_{ij} = \frac{\sum_{\alpha} w_{ij}^{\alpha} Y_{ij}^{\alpha}}{\sum_{\alpha} w_{ij}^{\alpha}}. \qquad (19)$$

The comparison graph is $G = \left(V, E\right)$ with $E = \{\{i, j\} : w_{ij} > 0\}$, and its 3-clique complex is $K_G^3 = \left(V, E, T(E)\right)$ with

$$T(E) = \{\{i, j, k\} \in \binom{V}{3} : \{i, j\}, \{j, k\}, \{k, i\} \in E\}. \qquad (20)$$

Cochain spaces: C^0 scores $s : V \to \mathbb{R}$, C^1 edge flows, C^2 triangular flows. The operators are

$$\left(\text{grad } s\right)\left(i,j\right) = (\delta_0 s)\left(i,j\right) = s_j - s_i, \qquad (21)$$

$$(\text{ curl } X)\left(i,j,k\right) = (\delta_1 X)\left(i,j,k\right) = X_{ij} + X_{jk} + X_{ki}, \qquad (22)$$

$$(\text{ div } X)(i) = -\left(\delta_0^* X\right)(i) = \textstyle\sum_{j:\{i,j\}\in E} w_{ij} X_{ij}, \qquad (23)$$

with the weighted inner product $\langle X,Y\rangle_w = \sum_{\{i,j\}\in E} w_{ij}X_{ij}Y_{ij}$ on C^1 and unweighted products on C^0, C^2. Closedness $\delta_{k+1}\delta_k = 0$ gives

$$\text{curl} \circ \text{grad} = 0, \qquad \text{div} \circ \text{curl}^* = 0, \qquad (24)$$

and the combinatorial Laplacians are

$$\Delta_k = \delta_k^*\delta_k + \delta_{k-1}\delta_{k-1}^*, \qquad \Delta_0 = \delta_0^*\delta_0 = -\text{ div grad }, \qquad \Delta_1 = \text{ curl}^* \text{ curl } - \text{ grad div }, \qquad (25)$$

One sign convention is used throughout, that of Jiang et al. 2011: $\Delta_0 = \delta_0^*\delta_0$ is the positive-semidefinite graph Laplacian $L = D - W$, and since Eq. (23) defines $\text{div} = -\delta_0^*$, this reads $\Delta_0 = -\text{ div grad}$. Eq. (27) below ($\Delta_0 s = -\text{ div } \bar{Y}$, $s^* = -\Delta_0^\dagger \text{div } \bar{Y}$) and the expression for Δ_1 are stated in that same convention. Δ_1 is the graph Helmholtzian, with $\dim \ker \Delta_k = \beta_k(K)$. The decomposition is

$$C^1 = \underbrace{\text{im}\left(\text{grad}\right)}_{\text{globally consistent}} \oplus \underbrace{\ker(\Delta_1)}_{\text{harmonic: globally cyclic}} \oplus \underbrace{\text{im}\left(\text{curl}^*\right)}_{\text{locally cyclic}}, \qquad \ker \Delta_1 = \ker(\text{ curl }) \cap \ker(\text{ div }), \qquad (26)$$

the three summands mutually orthogonal in $\langle\cdot,\cdot\rangle_w$. The least-squares potential and the residual split are

$$\min_{s\in C^0} \| \text{ grad } s - \bar{Y} \|_{2,w}^2 \Leftrightarrow \Delta_0 s = -\text{ div } \bar{Y}, \qquad s^* = -\Delta_0^\dagger \text{ div } \bar{Y}, \qquad (27)$$

$$R^* = \bar{Y} - \text{ grad } s^*, \qquad \Phi^* = \left(\text{curl curl}^*\right)^\dagger \text{curl } \bar{Y}, \qquad \text{proj}_{\ker \Delta_1} = I - \Delta_1^\dagger \Delta_1. \qquad (28)$$

Orthogonality turns the split into a genuine energy budget:

$$\|\bar{Y}\|^2 = \|\bar{Y}_{grad}\|^2 + \|\bar{Y}_{curl}\|^2 + \|\bar{Y}_{harm}\|^2, \qquad \rho_g = \frac{\|\bar{Y}_{grad}\|^2}{\|\bar{Y}\|^2},\ \rho_c = \frac{\|\bar{Y}_{curl}\|^2}{\|\bar{Y}\|^2},\ \rho_h = \frac{\|\bar{Y}_{harm}\|^2}{\|\bar{Y}\|^2}, \qquad (29)$$

with $\rho_g + \rho_c + \rho_h = 1$. Betti bookkeeping, with $B_1 \in \mathbb{R}^{|E|\times|V|}$ the matrix of δ_0 and $B_2 \in \mathbb{R}^{|T|\times|E|}$ that of δ_1:

$$\beta_0 = |V| - \text{ rank } B_1, \qquad \beta_1 = |E| - \text{ rank } B_1 - \text{ rank } B_2, \qquad \beta_2 = |T| - \text{ rank } B_2, \qquad (30)$$

checked against $\chi = |V| - |E| + |T| = \beta_0 - \beta_1 + \beta_2$. The flow is Construction A1, the Jaccard-normalized dominance

$$\bar{Y}_{ij}^{A1} = \frac{1}{|S_{ij}|}\sum_{a\in S_{ij}}\left(M_{ja} - M_{ia}\right) = \frac{c_j - c_i}{|\mathbf{m}_i \vee \mathbf{m}_j|}, \qquad S_{ij} = \{a : M_{ia} + M_{ja} \ge 1\}, \qquad (31)$$

whose pair-dependent denominator is what stops $\bar{Y}$ from being a pure gradient: with all primitives voting on every pair, $\bar{Y}_{ij} = \left(c_j - c_i\right)/d$ is exactly a gradient and $\rho_c = \rho_h = 0$ identically. Finally, the discrete circulation of a cycle $\sigma = (i_1 \to \dots \to i_L \to i_1)$,

$$W(\sigma) = \sum_{\ell=1}^{L} X_{i_\ell i_{\ell+1}}, \qquad (32)$$

vanishes on gradients, is contractible on curls, and on $\ker \Delta_1$ depends only on the homology class of σ — a real-valued circulation class, not a quantized topological charge.

Measurements

On yubiOS ($N = 2286$; symmetrized k-NN with $k = 8$ under Jaccard similarity; seed 20260812): $|E| = 16{,}438$, $|T| = 13{,}799$, and only 176 distinct rows of 512 possible. Measured,

$$\rho_g = 0.99961366, \qquad \rho_c = 0.00013059, \qquad \rho_h = 0.00025575, \qquad (33)$$

with $\beta_0 = 6$, $\beta_1 = 6756$, $\beta_2 = 6397$, $\text{rank } B_1 = 2280$, $\text{rank } B_2 = 7402$, and the Euler identity exact ($\chi = -353 = -353$). Numerical hygiene: $\langle \text{ grad , curl } \rangle / \| Y \|^2 = -5.4\times10^{-20}$, $\langle \text{ grad }, harm\rangle / \| Y \|^2 = -6.9\times10^{-16}$, $\langle \text{ curl }, harm\rangle / \| Y \|^2 = -1.1\times10^{-20}$, $\rho_g + \rho_c + \rho_h - 1 = 1.3\times10^{-15}$, B_2B_1 exactly zero, and a hexagon control returning $\beta_1 = 1$ — the last of these run out-of-band, since hodge_decompose.py exposes no callable hexagon entry point and proofs/VERIFY.sh substitutes a 20-node smoke decomposition in its place (proofs/PROOFS.md). On the cycle-18 corpus ($N = 1391$, same construction): $|E| = 10{,}236$, $|T| = 7149$, $\rho_g = 0.997422$, $\rho_c = 0.000841$, $\rho_h = 0.001737$, $\beta_1 = 5293$.

statistic	real	curveball-M null ($n = 50$)	z	degree-rewired G ($n = 50$)	z
ρ_{grad}	0.999614	0.999636 ± 0.000077	−0.29	0.974942 ± 0.000418	+59.04
ρ_{curl}	0.000131	0.000181 ± 0.000044	−1.12	0.001038 ± 0.000114	−7.98
ρ_{harm}	0.000256	0.000183 ± 0.000051	+1.39	0.024020 ± 0.000447	−53.14
β_1	6756	6561.7 ± 56.9	+3.41	$13,677.9 \pm 22.1$	−312.54

Table 7. Hodge quantities against two nulls: the matrix null (curveball on M, k-NN graph rebuilt from each drawn matrix) and the graph null (same M, same degrees, edges rewired). Every z is recomputed as (real − null mean)/null sd from the values printed in its own row, and each is attributed to its own ensemble: the +3.41 on β_1 is against the curveball-M null, the −312.54 against the degree-rewired null. One caution on the matrix-null column: because it rebuilds the graph per draw, its own real baseline is the rebuilt graph ($|E| = 16,465$, $\beta_1 = 6702$, $\rho_h = 0.00025453$) rather than the primary run's ($|E| = 16,438$, $\beta_1 = 6756$, $\rho_h = 0.00025575$) — a shift of about 0.2%. Against that rebuilt baseline the stored z for β_1 is +2.47; against the primary $\beta_1 = 6756$ printed in the "real" column here it is +3.41, and +3.41 is the figure used in the text. Source: results/B-hodge-yubios.json.

The reading is sharp. Against the *matrix* null every Hodge quantity sits within ~2.5σ: the coverage matrix's own marginals explain its entire Hodge profile. Against the *graph* null the corpus is dramatically less cyclic and less holey than chance. **The corpus is strongly anti-cyclic — it has far fewer independent holes than a random graph of the same shape.** This inverts rather than supports the Ginzburg–Landau vortex reading: harmonic mass is not concentrating in defect cores; there is less of it than chance, and it avoids the defect rows entirely.

Three pre-registered predictions, all falsified

P-H1 ("ρ_h collapses across $N_C \approx 40$–200, beyond the graph null") fails on its own stated criterion. With $k = 8$ fixed and 20 independent subsamples per N, ρ_h *rises* from 0.001405 ± 0.000270 at $N = 40$ to 0.003567 ± 0.000805 at $N = 80$ (a $+2.5\sigma$ increase), and at $N = 160$ (0.002412) is still above the $N = 40$ value. The monotone decline begins only after $N \approx 320$ and tracks edge density ($0.042 \to 0.0063$) rather than any critical band. What survives is the mechanism check, not the phenomenon: the rewired null rises monotonically ($0.0014 \to 0.0240$) where the real corpus falls, so the separation is real and enormous (z from +0.04 at $N = 40$ to −353.3 at $N = 2286$) — but it is smooth density-tracking, not a transition at N_C. This also removes the load-bearing status the pivot memo had assigned to P-H1's outcome.

P-H2 ("ρ_c tracks $1 - V_2$ with Spearman $\rho_s \geq 0.6$") fails with the wrong sign: across 11 corpora and subcorpora, $Spearman\left(1 - V_2, \rho_c\right) = -0.4545$, $p = 0.160$, and $Spearman\left(1 - V_2, \rho_h\right) = -0.4727$, $p = 0.142$. The curl maximum is docs ($N = 126$, $\rho_c = 0.011903$), the *smallest* corpus: ρ_c tracks graph density far more visibly than it tracks $1 - V_2$. The prediction's second clause — that the correlation should hold in the 24-D space the paper itself defined — is run here on a 24-D feature matrix rebuilt to ladder_correlations's own recipe (9 primitives, 12 NSS-like columns, 3 meta columns, seed 11) across eight sub-corpora and both NSS variants. It is **not supported**: point estimates +0.17 (independent NSS) and +0.10 (mass-correlated NSS), $p \geq 0.69$, indistinguishable from zero, with every configuration tried between −0.21 and +0.17. The power disclosure is owed in the same breath: at $n = 8$ a Spearman interval is roughly ±0.7 wide, so the test excludes only $\rho_s \gtrsim 0.7$ with confidence, not the +0.6 threshold itself. And in 24-D the Jaccard neighbour graph is dominated by the 12 random NSS columns, which carry no coverage signal — which makes the 24-D space a weak venue for the prediction rather than a fair one. That is a criticism of the clause, not a rescue of it: it is the space the paper defined, which is exactly why the clause should never have been posed there. Source: tests/T2-results.json.

P-H3 ("defect rows carry harmonic support, lift ≥ 2") fails by more than a factor of ten. With E_D the defect-incident edges, $lift = \left(\| h|_{E_D} \|^2 / \| h \|^2\right) / \left(|E_D|/|E|\right)$ is exactly 0.0 on the $k = 8$, $k = 16$ and $\tau = 0.35$ graphs (and 0.0 on the 1391-row corpus with its 14 strict defects), and 0.2054 on the coverage-band graph, against a calibration value of 0.17 for *randomly placed* zero rows. The mechanism is plain: the 204 zero-coverage rows are mutually identical, so k-NN links them only to each other, where $|\mathbf{m}_i \vee \mathbf{m}_j| = 0$ and the flow is identically zero — they form isolated components ($\beta_0 = 6$ at $k = 8$, 221 on the threshold graph). Clause (ii), quantized azimuthal S^2 winding, is neither confirmed, falsified nor untested: it is **void**. The clause asked whether harmonic 1-cocycles are 2π-quantized, but the harmonic component is a real-valued flow in Jaccard-normalized dominance units, in which 2π is not a unit of anything; the clause is dimensionally malformed, and its winding half is unfalsifiable by construction (Section 6.5). The empirical part is nonetheless run and reported. Building the S^2 embedding by the paper's own pipeline (z-score, PCA top-2, RMS rescale, stereographic lift; $\rho_h = 2.5575 \times 10^{-4}$, $\beta_1 = 6756$, $|E| = 16,438$ — identical to the primary run) and taking the 14,158 fundamental cycles of the $k = 8$ graph, harmonic circulation is *not* quantized: every one of the 14,158 cycles has $round\left(\oint h / 2\pi\right) = 0$, the largest circulation in the complex is $|\oint h| = 0.1010$, i.e. 1.6% of 2π, and the distribution is a single blob near zero rather than a ladder of integer multiples. (The real corpus's mean distance to the nearest 2π multiple exceeds the curveball null's at $z = +14.97$, but that is one test and not two — "mean distance of $C/2\pi$ to an integer" and "mean $|C|$" are the same statistic divided by 2π — and it is a magnitude effect, not a lattice effect: unnormalized circulations are $\approx 9\times$ larger than the null's, while the scale-invariant harmonic energy ρ_h is *not* significant, $z = +1.45$, $p = 0.157$. All null p-values in this block come from 50 curveball draws and are therefore resolution-limited at $p < 0.02$; the Rayleigh Z is referred to its empirical rank only and is not

z-scored, being a non-Gaussian, n-dependent statistic.) The one non-trivial thing the winding computation yields is the winding-number histogram: 86.4% of cycles have $W = 0$ and none exceeds ± 2, so the embedding's cycles almost all fail to enclose the pole. The apparent correspondence between circulation and winding ($r\left(|C|, |W|\right) = +0.95$) is a binary-degeneracy artifact — $|C|$ takes essentially two values, 0 and 0.100954 — and the rank correlation, which is not fooled by it, is $\rho_s = +0.31$. Source: tests/T1-results.json.

Reporting three falsifications from three pre-registered predictions is the methodology working, not the channel failing. The falsifications are what license the replacement claim (anti-cyclicity), which was not predicted in advance and would have been unavailable without the graph null.

Construction invariance

construction	$\lvert E\rvert$	β_0	β_1	ρ_g	ρ_c	ρ_h	$z\left(\rho_c\right)$	$z\left(\rho_h\right)$	$z\left(\beta_1\right)$
k-NN $k = 8$	16,438	6	6756	0.999614	0.000131	0.000256	−7.98	−53.14	−312.5
k-NN $k = 16$	32,068	2	7222	0.998833	0.000542	0.000624	−22.09	−62.68	−298.8
threshold $\tau = 0.35$	20,000	221	13,286	0.999441	0.000141	0.000418	−1.91	−19.12	−16.1
coverage-band $b = 1$	20,000	1	14,886	0.990753	0.001099	0.008148	+1.29	−26.34	−37.3

Table 8. Four graph constructions on the same M, each against its own degree-rewired null. The $\tau = 0.35$ and coverage-band rows both sit exactly at the builder's $|E| = 20,000$ edge cap (field edge_cap in the source JSON; the two k-NN rows are uncapped at $60,000$), so their β_0 and β_1 are in part cap artifacts — which is worth stating, since these two rows carry the $32\times$ spread in ρ_h and the sign flip of the ρ_c verdict. Source: results/D-construction-sensitivity.json.

Construction-invariant, and therefore reportable as corpus properties: (i) $\rho_g > 0.99$ everywhere — the flow is dominated by a potential and the corpus is near-totally ordered by coverage; (ii) ρ_h far below the degree-matched null ($z \in \left[-63, -19\right]$); (iii) β_1 far below it ($z \in \left[-313, -16\right]$); (iv) defect lift $\ll 2$ (max 0.205); (v) orthogonality residuals $< 10^{-15}$ and the Euler identity satisfied in every run. *Not* invariant, and therefore not reportable as corpus scalars: the magnitude of ρ_h (spans $32\times$, $0.000256 \to 0.008148$), of β_1 ($2.2\times$) and β_0 ($1 \to 221$); the sign of the ρ_c verdict (below null at k-NN, indistinguishable at threshold, above null at coverage-band); and harmonic-support concentration (26 edges carry 50% and 369 carry 90% of harmonic energy at k-NN, of $16,438$, with only 712 of $14,158$ fundamental cycles carrying non-zero circulation, whereas on the coverage-band graph the support is diffuse — 672 edges for 50% and $12,623$ of $17,715$ cycles non-zero).

The standard-candle generator and calibrated detection

Everything above is negative or corrective. The positive result is an instrument: a generator of corpora whose curvature is prescribed by construction, so that a detection statistic can be calibrated rather than asserted.

Generator

Fix N, d, a base rate p_0, a mode count m, and an amplitude $s \ge 0$. Lay down the Fibonacci lattice of Eq. (9), evaluate the 16 real spherical harmonics of $L = 3$ at each point x_i, and take the first m probe channels $g_1, \ldots, g_m$ in the fixed order Y_3^3, Y_3^{-3}, Y_2^2, Y_2^{-2}, Y_3^0 (channel 0 is always Y_3^3). Standardize each channel,

$$\tilde{g}_k(x_i) = \frac{g_k(x_i) - \bar{g}_k}{sd\left(g_k\right)}, \qquad (34)$$

which is what makes the construction invariant to the normalization constant K of Eq. (11). Assign deterministic column loadings w_{kj}, column j carrying channel j mod m with sign alternating by block, so that distinct columns co-vary *through the planted curve* rather than merely through row mass. Then

$$\eta_{ij} = \mu + s\sum_{k=1}^{m} \tilde{g}_k(x_i)\, w_{kj}, \qquad \mu = \text{logit}\left(p_0\right), \qquad (35)$$

$$X_{ij} \sim Bernoulli(\sigma\left(\eta_{ij}\right)), \qquad \sigma(u) = \frac{1}{1 + e^{-u}}. \qquad (36)$$

The amplitude s, in log-odds units, is the ground truth. Two further kinds are generated for reference: mixture, a k-class latent mixture with random column-probability centres, and null, iid Bernoulli at the column marginals of the matched $s = 0$ corpus.

Detection

Procedure 1 (calibrated detection).

Given a binary corpus X of shape $N \times d$: (1) compute V_2 by Eq. (5); (2) draw R matrices from the curveball ensemble at X's own N, d and marginals; (3) report $\Delta V_2 z$ by Eq. (13) *together with* R, since z is rep-count sensitive at the 15–20% level; (4) reject the matched-marginal null M_0 when $\Delta V_2 z > 1.645$. Procedure 1 is a definition, not a result. What follows is its measured behaviour on corpora whose amplitude s is known by construction — an empirical characterization on one generator family and one grid, with no claim of generality beyond it.

Empirical Result 1 (measured detection behaviour). Let X be generated by Eqs. (34)–(36) at amplitude s and passed through Procedure 1. Over the grid of Table 9 — one generator family, one base rate, $m = 2$ modes, 8 independent corpora per cell at $N \in \{128, 512\}$ and 4 at $N = 2048$, each against its own curveball ensemble — the following is measured, not proved:

- *Detection power is monotone non-decreasing in* s *over* $s \geq 0.5$ *at every* $\left(N, d\right)$ *in the grid, and strictly increasing for* $N \geq 512$. At $N = 512$, $d = 9$ the power at $z > 1.645$ runs $0.00 \to 1.00 \to 1.00$ across $s = 0.5, 1, 2$; at $N = 512$, $d = 16$, $0.25 \to 1.00 \to 1.00$; at $N = 2048$, $d = 9$, 1.00 throughout; at $N = 2048$, $d = 16$, $0.50 \to 1.00 \to 1.00$. Cell-mean $\Delta V_2 z$ is likewise increasing in s over the same range at $N \geq 512$ ($+0.04 \to +3.43 \to +12.86$ at $N = 512$, $d = 9$; $+2.46 \to +10.82 \to +28.07$ at $N = 2048$, $d = 9$), and increasing in N at fixed $s \geq 1$ (+0.95, +3.43, +10.82 at $s = 1$, $d = 9$, $N = 128, 512, 2048$).

- *At* $s = 0$ *the statistic is centred at zero and over-dispersed, and the false-positive rate is bounded but above nominal.* On 48 fresh $s = 0$ corpora at $N = 512$, $d = 9$: mean $z = -0.014$, sd 1.320 where a calibrated z would give 1.000; one-sided $FPR = 0.083$ at the nominal-5% threshold $z > 1.645$, two-sided 0.104 at $|z| > 1.96$. The over-dispersion is disclosed here and carried through Section 8: every z in this paper should be read as inflated by roughly a third in units of sd.

- *Monotonicity fails at low power, and no ordering is claimed there.* At $N = 128$, $d = 9$ the cell means are $+0.59 \pm 0.36$ ($s = 0$), $+0.63 \pm 0.33$ (0.25), $+0.09 \pm 0.65$ (0.5), $+0.95 \pm 0.66$ (1.0) — unordered, and every one within Monte-Carlo error of zero at 8 reps per cell (± is the standard error of the cell mean). The same happens at $s = 0.25$ wherever power sits on its floor: -1.04 ± 0.42 at $N = 2048$, $d = 9$, *below* the $+0.32 \pm 0.53$ of $s = 0$ in the same column, and at $N = 512$, $d = 9$ the $s = 0.25$ cell records power 0.12 against 0.00 at $s = 0.5$ — one of eight corpora crossing threshold by chance, which is what an 8.3% false-positive rate looks like at eight replicates. Cell means at $s \leq 0.25$, and all cell means at $N = 128$, must not be read as ordered.

- *The sensitivity limit is* $s \approx 1$ *log-odds at* $N \leq 512$. At $N = 512$, $d = 9$: $s = 0.5$ gives +0.04 and power 0.00; $s = 1.0$ gives +3.43 and power 1.00. A non-detection below that amplitude is a statement about power, not about the world.

- *Raw* V_2 *has none of these properties*: it orders corpora by $\left(N, d\right)$ rather than by s. $V_2 = 0.3083$ at $\left(N = 128, d = 9, s = 0\right)$ exceeds $V_2 = 0.3077$ at $\left(N = 2048, d = 9, s = 2\right)$, while $\Delta V_2 z$ separates the same pair as +0.59 against +28.07.

None of the above is a theorem, and it is deliberately not presented as one: there is no hypothesis set that survives outside this grid, every clause is design-dependent, and the thresholds do not transfer across N, d, base rate, or mode count (Section 8). Presenting a Monte-Carlo sweep in a theorem environment is precisely the failure this paper indicts elsewhere in its own lineage. All numbers in (i)–(v) are from results/signal_recovery.json (cell means and standard errors recomputed from its 248 per-corpus rows) and results/SUMMARY.md §2.

N	d	s	**reps**	V_2	V_2 **null**	ΔV_2	$\Delta V_2 z$	**power**	R^2_{paper}	R^2_{cv}
128	9	0.0	8	0.3083	0.3019	+0.0064	+0.59	0.12	1.0000	0.0568
128	9	0.25	8	0.3079	0.3004	+0.0076	+0.63	0.25	1.0000	0.0487
128	9	0.5	8	0.3010	0.3003	+0.0007	+0.09	0.25	1.0000	0.0554
128	9	1.0	8	0.3098	0.3002	+0.0096	+0.95	0.25	1.0000	0.1354
128	9	2.0	8	0.3428	0.3013	+0.0415	+3.69	0.88	1.0000	0.0944
128	16	2.0	8	0.2361	0.1984	+0.0377	+5.34	1.00	1.0000	0.0612
512	9	0.0	8	0.2587	0.2603	−0.0016	−0.27	0.00	1.0000	0.0928
512	9	0.5	8	0.2605	0.2604	+0.0001	+0.04	0.00	1.0000	0.1063
512	9	1.0	8	0.2783	0.2615	+0.0168	+3.43	**1.00**	1.0000	0.0973
512	9	2.0	8	0.3250	0.2613	+0.0637	+12.86	1.00	1.0000	0.2033
512	16	1.0	8	0.1811	0.1602	+0.0209	+6.61	1.00	1.0000	0.0940
512	16	2.0	8	0.2236	0.1612	+0.0624	+20.32	1.00	1.0000	0.1432
2048	9	0.0	4	0.2415	0.2407	+0.0007	+0.32	0.00	1.0000	0.1152
2048	9	0.5	4	0.2469	0.2412	+0.0057	+2.46	1.00	1.0000	0.1067
2048	9	1.0	4	0.2712	0.2411	+0.0301	+10.82	1.00	1.0000	0.1010
2048	9	2.0	4	0.3077	0.2423	+0.0654	+28.07	1.00	1.0000	0.1702
2048	16	1.0	4	0.1689	0.1422	+0.0267	+18.68	1.00	1.0000	0.1133
2048	16	2.0	4	0.2266	0.1428	+0.0838	+57.67	1.00	1.0000	0.1942

Table 9. Signal recovery and power (selected rows of the 30-cell sweep; "power" is the detection rate at $z > 1.645$). R^2_{paper} is the predecessor's sphere-fit statistic: 1.0000 in every cell, at every amplitude, including $s = 0$. Sources: results/signal_recovery.json, results/SUMMARY.md §2.

Fold-aggregated detection

Clause (iii) of Empirical Result 1 is a statement about *cells*. At $N = 128$ each amplitude is measured alone, its $\Delta V_2 z$ sits at the power floor, and the four cell means are not ordered. Nothing in that clause says the *trend* is unavailable — only that no single cell carries it. This subsection tests the trend itself.

Constant-ratio ladder (algebraic fold). Fix a base amplitude s_0 and a ratio $\rho > 1$, and generate the family

$$s_k = s_0\, \rho^{k}, \qquad k = 0, \ldots, K-1, \qquad (37)$$

every rung emitted by the generator of Eqs. (34)–(36) at one fixed $\left(N, d, p_0, m\right)$. Here $s \in \{0.25, 0.5, 1.0, 2.0\}$, so $\rho = 2$ and $\log_2 s$ is the integer grid $-2, -1, 0, +1$. The construction is taken from a reader-supplied "ratios-of-ratios" spreadsheet whose blocks hold b/a constant along ladders terminating at $384 = 2^7 \cdot 3$ (factor pairs 8×48, 12×32, 16×24); we borrow only the defining property — constant ratio between consecutive rungs — and call such a family an *algebraic fold (constant-ratio amplitude family)*.

Paired design. A ladder is *paired* when one generator seed is drawn per trial and reused at every rung, so the K corpora of a trial share a common Bernoulli draw stream and differ only in s; *unpaired* when a fresh seed is drawn at every $\left(\text{trial}, \text{rung}\right)$. Pairing is a variance-reduction device on the within-trial trend and changes no cell's expectation; measured here it cuts the variance of the fold slope by about $2.4\times$.

Fold-slope statistic. Within trial t, regress the K measured z values on $\log_2 s$ and keep the ordinary least-squares slope; across T trials report its t-statistic:

$$\hat{\beta}_t = \frac{\sum_k (\log_2 s_k - \overline{\log_2 s})(z_{t,k} - \bar{z}_t)}{\sum_k (\log_2 s_k - \overline{\log_2 s})^2}, \qquad t_{fold} = \frac{\bar{\hat{\beta}}}{SD\left(\hat{\beta}\right)/\sqrt{T}}. \qquad (38)$$

The reference distribution is *not* Gaussian theory — the per-cell z is itself over-dispersed (Empirical Result 1(ii), $sd = 1.320$ at $s = 0$) — but the empirical *null ladder*: the identical procedure with $s = 0$ at all K rungs, whose 95% quantile of $\hat{\beta}$ is the critical value.

Empirical Result 2 (fold-aggregated detection).

At $N = 128$, $d = 9$, $m = 2$, $T = 30$ trials and 30 curveball replicates per cell, on the ladder of Eq. (37) with $\rho = 2$: (i) the per-cell picture of Empirical Result 1(iii) is unchanged — the two low rungs sit at the power floor (0.067 and 0.100 detection at $z > 1.645$) and their means are within Monte-Carlo error of each other; (ii) per-trial monotonicity across the four rungs holds in only 0.567 (paired) and 0.633 (unpaired) of trials, so the ladder is *not* monotone trial by trial either; (iii) nevertheless the fold slope of Eq. (38) is enormous and its null is tight — $t_{fold} = 45.19$ paired, 28.92 unpaired, against a null-ladder slope of -0.026 with 95% quantile 0.0997 — giving fold-slope power 1.00 in both designs. Table 10 is the full record.

rung s	$\log_2 s$	mean z (paired)	mean z (unpaired)	power ($z > 1.645$)
0.25	−2	−0.055	+0.066	0.067
0.50	−1	+0.478	+0.645	0.100
1.00	0	+6.664	+6.504	1.00
2.00	+1	+20.657	+20.536	1.00
trial-level aggregates, $T = 30$ trials				
monotone-trial fraction	—	0.567	0.633	—
fold slope t_{fold}, Eq. (38)	—	**45.19**	**28.92**	—
fold-slope power vs null $q_{95} = 0.0997$	—	**1.00**	**1.00**	—

Table 10. Fold-aggregated detection on the constant-ratio ladder $s = 0.25, 0.5, 1, 2$ at $N = 128$, $d = 9$, $m = 2$; $T = 30$ trials, 30 curveball replicates per cell. Pairing means a common generator seed across the rungs of a trial; it reduces the variance of $\hat{\beta}$ by about $2.4\times$ (hence the higher t at an essentially unchanged slope) and leaves every cell mean where it was. The null ladder ($s = 0$ at all four rungs, same T) has slope mean −0.026 and 95% quantile 0.0997; that quantile, not a Gaussian tail, is the critical value. Source: results/fold-ladder-experiment.json, regenerated by scripts/fold_ladder_experiment.py.

What is and is not repaired. Repaired, at the family level: the non-monotone clause (iii) of Empirical Result 1 no longer forces "no ordering is available at $N = 128$". A constant-ratio family is one experiment, and as one experiment it is monotone and fully powered. Not repaired, at the cell level: nothing here raises the power of an individual low-amplitude cell. The $s = 0.25$ and $s = 0.5$ rungs still detect at 0.067 and 0.100; per-trial monotonicity is still a coin flip; the information floor of Empirical Result 1(iv) is untouched. Fold aggregation buys a *trend* test, not a *cell* test, and the two must not be quoted for each other. The fold-slope false-positive rate is controlled by the null-ladder quantile and by nothing else (Section 8).

The sphere-fit R^2 is degenerate; its cross-validated form is not

The last two columns of Table 9 carry a second correction. The predecessor's sphere-fit R^2 returns 1.0000 on everything — planted corpora at every amplitude, and independent-column noise — because the $L = 1$ spherical-harmonic block is linear in (x, y, z) and the fit is evaluated in-sample on points constructed from the very embedding it is fitting. It is not a measure of curvature; it is an identity. The cross-validated $L \geq 2$ variant *is* informative: on a GWTC-like corpus at $N = 1000$ it returns $R^2_{cv} = 0.7678$ against 0.1391 on that corpus's column-permuted null (at $N = 138$: 0.6148 against 0.0175). The same pair in V_2: 0.6624 with $z = +111.3$ and proxy $\hat{r} = 3.02$ for the real generator, against 0.2495 with $z = -0.13$ and $\hat{r} = 8.02$ for its null (both $\hat{r} = 2/V_2$, Eq. (15)). The false-positive block confirms that $z\left(\Delta R^2_{cv}\right)$ is conservative rather than anti-conservative: mean -0.265, sd 0.884, one-sided $FPR = 0.021$.

The generator's own calibration pack ($N = 200$, $d = 9$, $m = 2$, 8 trials per amplitude, 60 null reps) separates statistic from diagnostic in one table: $\Delta V_2 z$ sweeps $+0.18$, $+0.10$, $+1.50$, $+5.49$, $+10.28$, $+19.34$, $+28.01$ across amplitudes 0, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0, reaching power 1.00 at $s = 0.75$, while the sphere R^2 stays flat at ≈ 0.48 across the entire sweep and the gate $PC1 + PC2$ passes throughout. Detection threshold: $s = 0.75$ at this $\left(N, d, p_0, m\right)$; FPR at $|z| > 3$ is 0.00, at $|z| > 2$ is 0.125. A worked single corpus at $s = 1.5$: $V_2 = 0.4477$, participation ratio $PR = 6.837$ (Eq. (14); the two-share proxy on the same corpus is $\hat{r} = 2/0.4477 = 4.47$), sphere $R^2 = 0.4330$, $PC1 + PC2 = 0.4474$ (gate passes), design numerical rank 16, condition number 1.006, curveball null 0.2864 ± 0.0091, $\Delta V_2 = +0.1613$, $\Delta V_2 z = +17.67$ *at* 200 *null reps*. The gate passing at 0.4474 says little; the load-bearing line is +17.67. The same corpus is measured again in Section 6 at 60 null reps and scores +20.62: z is rep-count sensitive at the 15–20% level, which is why Procedure 1 requires the rep count to be reported beside every z.

Parseval energy shares

The sphere fit has so far been read through one scalar, R^2, which Section 5 showed to be degenerate in-sample. The design it is computed on carries more than that. Under the Fibonacci sampling of Eq. (9) the 16 real spherical harmonics of $L = 3$ are *near-orthonormal at the sampled points*: the Gram matrix satisfies

$$B^{\top} B \approx \frac{N}{4\pi} I_{16}, \qquad (39)$$

with measured condition number 1.003–1.006 and numerical rank 16 on every design in this paper (the worked planted corpus of Section 5: 1.006; the real GWTC design of Section 6.4: 1.0034). Equation (39) is what makes an energy budget available: writing the ridge solution of Eq. (4) as C with rows indexed by mode (ℓ, m), the fitted energy partitions across modes,

$$\| \hat{Z} \|_F^2 = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} \| B_{(\ell,m)} \|^2 \, \| C_{(\ell,m)} \|^2, \qquad E_{\ell m} = \frac{\| B_{(\ell,m)} \|^2 \, \| C_{(\ell,m)} \|^2}{\sum_{\ell' m'} \| B_{(\ell',m')} \|^2 \, \| C_{(\ell',m')} \|^2}, \qquad (40)$$

so $\sum_{\ell m} E_{\ell m} = 1$ exactly and the per-degree aggregate $E_\ell = \sum_m E_{\ell m}$ is a point on the 3-simplex. This is Parseval's identity for the sampled design, not an approximation: the residual of the partition is the 0.3%–0.6% by which Eq. (39) fails to be exact.

Two properties recommend E as a coordinate of the map Φ. First, it is *dimension-comparable*: E has the same 16 entries whatever N and d are, unlike V_2, which Proposition 1 shows is a function of D. Second, it is *scale-free along a fold*: multiplying an amplitude by ρ scales numerator and denominator of Eq. (40) together, so E is invariant along a constant-ratio ladder of Eq. (37) up to the change in the fitted *shape*. This is the same algebra that made the legacy-K error of Eq. (12) scale-only: a common factor on a standardized channel cancels in every ratio built from it.

We therefore propose the per-degree shares and the three-fold azimuthal share as additional coordinates,

$$s_{Parseval}(M) = [E_0, E_1, E_2, E_3, E_{3,3}], \qquad (41)$$

each standardized against its family by Eq. (43) exactly as the coordinates of Eq. (42) are. $E_{3,3}$ is singled out because Y_3^3 is the probe channel the whole program has been asking about; under Eq. (40) it becomes a *share of a fixed budget*, so it is finally comparable across corpora, and — being null-standardizable like everything else — finally falsifiable. Section 6.4 is its first application, and the answer there is negative.

The IS-THIS-X question space

Definition

Definition 1 (question space). *The question space is the triple $\left(\Phi, \mathcal{M}, \mathcal{V}\right)$, where Φ maps a corpus to a null-standardized statistic vector, $\mathcal{M}$ is an extensible set of reference families generated at the observed $\left(N, d, \text{density}\right)$, and $\mathcal{V}$ is an exclusion-only verdict rule.*

The statistic vector is

$$s(M) = [V_2, \Delta V_2 z, PR, Otsu\left(\text{row-mass histogram}\right), \bar{\pi}, sd(\pi)], \qquad \pi_a = \frac{1}{N} \sum_i M_{ia}, \qquad (42)$$

with PR the participation ratio of Eq. (14) — not the two-share proxy $\hat{r}$, which would be a deterministic function of the first coordinate and would add a spurious dimension to the question space — optionally extended by the Hodge block $\left[\rho_g, \rho_c, \rho_h, \beta_1\right]$ under a fixed, declared graph construction. Every coordinate is invariant under row relabeling; the marginal coordinates are equivariant under column permutation and are summarized by permutation-invariant moments. The invariance requirement is not decoration: it deletes the entire row-lag κ family by construction, since a statistic computed along file-listing order cannot be a property of a corpus. A second membership condition is imposed in Section 6.5 and is of the same kind: no coordinate is admitted without a demonstrated non-degenerate null, a construction under which the statistic provably *could* have taken a different value.

Each coordinate is standardized against each family's own mean and sd, with a 5% relative SD floor so a noisily narrow family cannot exclude its own members:

$$z_{f,c}(M) = \frac{s_c(M) - \mathbb{E}_f[s_c]}{\max\left(SD_f[s_c],\ 0.05\,|\mathbb{E}_f[s_c]|\right)}. \tag{43}$$

The verdict rule is exclusion-only:

$$\mathcal{V}\left(M, f\right) = \begin{cases} \texttt{excluded} & \text{if } \max_c |z_{f,c}(M)| > 3, \\ \texttt{not-excluded} & \text{if } \max_c |z_{f,c}(M)| \le 3, \\ \texttt{not-tested} & \text{if } f \text{ is degenerate on every coordinate.} \end{cases} \tag{44}$$

PARTIAL and "compatible with" are not permitted outputs. The default families are M_0 (iid Bernoulli at matched column marginals — the origin), M_1^{weak} and M_1^{strong} (planted curve at $s = 0.5$ and $s = 1.5$), and M_4 (two-class latent mixture at separation 1.5). The set is extensible, and the correct response to an all-excluded result is to widen the grid, not to relabel the nearest family.

Round-trip validation

The map is trustworthy only if a corpus of known provenance recovers its own family and excludes the origin. On the planted corpus at $N = 200$, $d = 9$, $s = 1.5$ — the same corpus worked through in Section 5, here measured at 60 null reps rather than 200, hence $\Delta V_2 z = 20.62$ against the +17.67 quoted there — the measured vector is

$$s = \left[\,V_2\,0.4477,\ \ \Delta V_2 z\,20.62,\ \ PR\,6.837,\ \ Otsu\,0.6976,\ \ \bar{\pi}\,0.5139,\ \ sd(\pi)\,0.0343\,\right], \tag{45}$$

with verdicts: M_0 excluded at $\max|z| = 27.91$ (driving coordinate $\Delta V_2 z$); M_1^{weak} excluded at 15.15 ($\Delta V_2 z$); M_1^{strong} not-excluded at $\max|z| = 0.99$ (driving coordinate PR); M_4 excluded at 3.89, driven by column-marginal spread rather than by V_2 (where $z = 2.07$) — which is precisely the coordinate on which curve and mixture genuinely overlap. A matched null corpus returns $\Delta V_2 z = -0.116$ and does not exclude M_0. Own family recovered at $\max|z| < 1$, origin excluded at $z \approx 28$: that round trip is what licenses reading the map on real corpora.

Placement of yubiOS and of the GWTC-like corpus

family	**driving coordinate**	**evidence**	**verdict for yubiOS** (2286×9)
M_0 matched-marginal null	$\Delta V_2 z$	+12.13	excluded
M_2 mean-field Ginzburg–Landau	shape of $\Delta V_2(N)$	flat, no interior max	excluded
M_4 two-population mixture	V_2, PR, Otsu	nearest family	not-excluded
M_6 Hodge / topological (cyclicity)	ρ_h, β_1 vs matrix null	+1.39, +3.41	not-excluded (no excess cyclicity)
M_1 planted curve	$\Delta V_2 z$ magnitude	+12 vs +20 to +28 at $s \ge 1.5$	not-tested at matched $\left(N, d\right)$

Table 11. Placement of yubiOS. The corpus sits off the origin, but only just: the nearest family is a two-population mixture, with a small positive residual $\Delta V_2 = +0.0144$.

yubiOS is M_0-excluded at $z = +12.3$, nearest a two-population latent-class mixture with a small positive residual. That one sentence subsumes the cross-corpus work of cycles 27–34: corpora with unimodal row mass sit near the origin (docs, $z = -0.10$), and corpora that pool a saturated population with a sparse one sit far from it (whole corpus $z = +12.1$; SCAP SSG $z = +21.5$). Mean-field Ginzburg–Landau is excluded on the shape of $\Delta V_2(N)$ (Table 4), not on the sign of a single z — which matters, because the sign of that single z is exactly what the retraction of Remark 2 reversed. The generated latent-class series reproduces yubiOS's signature: a two-class mixture shows a small constant ΔV_2 whose z grows only with N (+0.9 at $N = 40$ to +7.7 at $N = 4096$), while a one-class corpus stays at $|z| \le 1.5$ throughout.

The GWTC-like corpus places elsewhere, and is the useful contrast: $V_2 = 0.6624$, $\Delta V_2 z = +111.3$, proxy $\hat{r} = 3.02$, $R^2_{cv} = 0.7678$ at $N = 1000$, against its own column-permuted null at $V_2 = 0.2495$, $z = -0.13$, $R^2_{cv} = 0.1391$; at $N = 138$, $V_2 = 0.6722$, $z = +34.3$. The two corpora are not near neighbours in s-space — which is the honest form of the cross-domain claim earlier cycles tried to make by narrative. It must also be said that a continuous physical-parameter matrix and a binary coverage matrix are not the same object: the GWTC-like corpus enters this map only through a stated binarization, and any claim about real gravitational-wave catalogues would need real posterior samples under that same stated rule.

Real-catalog application: GWTC

The GWTC-like corpus above is synthetic, and the caveat closing the previous subsection — that a claim about real gravitational-wave catalogues needs real data — is discharged here rather than deferred again. The GWOSC eventapi GWTC superset was fetched on 2026-08-12: 391 events, of which 269 carry all nine of mass_1_source, mass_2_source, chirp_mass_source, total_mass_source, luminosity_distance, chi_eff, redshift, final_mass_source, network_matched_filter_snr, spanning GWTC-2.1-confident (53), GWTC-3-confident (35), GWTC-4.1 (84) and GWTC-5.0 (97). Luminosity distance is taken in log; every column is then z-scored. The matrix is continuous, so the fixed-margin curveball null is inapplicable by construction and the matched null is the independent per-column shuffle — which also means this corpus stays *off* the binary IS-THIS-X map of Section 6 unless and until it is binarized under a stated rule, exactly as the skill's constraint requires.

quantity	value	null / comparison
$V_2 = PC1 + PC2$ (269×9)	0.835294	column-shuffle 0.275263 ± 0.008355 (200 reps), $z = +67.0$
PC1 / PC2	0.6635 / 0.1718	—
correlation eigenvalues	5.9716, 1.5461, 0.9729, 0.3831, 0.0867, 0.0383, 0.0010, 0.0004,	≈ 0
sphere fit R^2	0.7992	chordal residual mean 0.1519, p95 0.3680
design condition / numerical rank	1.0034 / 16	Eq. (39)
Parseval per-degree shares E_ℓ	$E_0 = 0.0474$, $E_1 = 0.6464$, $E_2 = 0.2279$, $E_3 = 0.0782$	
Y_3^3 share $E_{3,3}$	0.00334	null 0.00832 ± 0.00427, $z = -1.17$ — **not significant**
synthetic GWTC-like, $N = 1000$ (prior)	$V_2 = 0.6624$	$z = +111.3$, $R^2_{cv} = 0.7678$ (results/gwtc.json)

Table 12. The real GWTC catalogue on the continuous branch of the pipeline. The last row is the synthetic GWTC-like corpus of Section 5, retained for comparison: cycle-34's "GWTC" number ($V_2 = 0.4003$) and the 0.6624 quoted above it were both synthetic-mixture constructions, and are superseded — not contradicted — by the real-catalog row. Sources: results/real-gwtc-results.json, data/real/real-gwtc-matrix.csv, data/real/gwtc-real-gwosc.json; regenerated by scripts/real_gwtc_pipeline.py.

The redundancy disclosure comes first. The eigenvalue spectrum of Table 12 ends 0.0010, 0.0004, ≈ 0: the 9×9 correlation matrix is numerically rank-deficient, because three of the nine fields are known algebraic functions of two others. Chirp mass, total mass and final mass are determined by $\left(m_1, m_2\right)$ up to a small radiated-energy correction, and redshift and luminosity distance are related by the cosmology. A V_2 of 0.835 at $z = +67$ therefore measures, in large part, *kinematic identities that were true before any data were taken*. This is not a discovery about black-hole populations; it is the map correctly reporting that the coordinate system is redundant. The honest reading of $z = +67$ is "the columns are massively dependent", and the interesting question — whether structure survives after the identities are quotiented out — is not answered here.

The Y_3^3 result is a null, and is reported as one. Under Eq. (40) the real catalogue puts 0.334% of its fitted energy in the three-fold azimuthal mode, against a column-shuffle null of $0.832 \pm 0.427\%$: $z = -1.17$, comfortably inside the null band and, if anything, on the low side. Thirty-four cycles of pentagon and three-fold-symmetry narrative do not survive their first contact with a real catalogue under a matched null. The energy that is there is at $\ell = 1$ (64.6%, dominated by the $\left(1, 1\right)$ and $\left(1, -1\right)$ modes at 0.402 and 0.158) and $\ell = 2$ (22.8%) — the smooth, low-order shape of a two-dimensional principal plane, which is what the V_2 and sphere-fit numbers already said.

Synthetic to real. The prior GWTC rows remain in results/gwtc.json and in Table 1; they are a generator's output and were never a measurement of the sky. What changes with this subsection is only that the map has now been run once on a real catalogue, on the continuous branch, with its own matched null — and that the one azimuthal claim it could have supported, it does not.

Identity checks: coordinates that cannot fail

Five quantities in this program's lineage are true by construction, and each was at some point read as evidence. They are collected here because the pattern, not any one instance, is the finding.

quantity	status	why it carries no data
$V_2 \geq 0.40 \Leftrightarrow \hat{r} \leq 5$	identity	$\hat{r} \equiv 2/V_2$ (Prop. 2)
sphere-fit $R^2 = 1.0000$	identity	$L = 1$ block linear in $\left(x, y, z\right)$, evaluated in-sample on the embedding it fits
$\sum wrap\left(\Delta\varphi\right) \in 2\pi\mathbb{Z}$	identity	wrapped increments of any angle-valued field sum to $2\pi\mathbb{Z}$ around any closed loop; measured deviation 4.4×10^{-16} on all 14,158 cycles
Kolmogorov cycle criterion on the edit log	vacuous	zero backward edges make the chain a DAG; no directed cycle exists to test (§7)
$\delta\left(k\right) = \Phi\left(k\right) - \Phi\left(k+1\right)$	identity	the ladder distance is a fixed function of the coverage count; the gradient property follows from the definition (§7, D3)

Table 13. Vacuous-by-construction quantities found in this program. Each was reported at least once as a substantive result.

The design rule this imposes on the map of Definition 1 is a membership condition, and it is cheap: **no coordinate is admitted without a demonstrated non-degenerate null** — a construction under which the statistic provably *could* have taken a different value. Three of the five above fail that test instantly: $\hat{r}$ is a function of V_2, in-sample R^2 is a function of the design, and the winding sum is a function of the wrapping convention. Two fail it subtly, and are the instructive ones: a criterion can be satisfied vacuously by the very extremity of the effect it was meant to detect (the DAG), and a conservation law can hold identically because the observable is a function of the state variable alone (the ladder). P-H3 clause (ii) is the sharpest case: it asked whether harmonic 1-cocycles are 2π-quantized, but the harmonic component is a real-valued flow in Jaccard-normalized dominance units, in which 2π is not a unit of anything. The clause was not merely false; it was dimensionally malformed, and the winding half of it was unfalsifiable. A pre-registered prediction can fail in this fourth way — neither confirmed, falsified, nor untested, but *void* — and a question space that issues exclusion-only verdicts must be able to say so.

A second channel: Φ_{dyn} and the family M_{dyn}

Section 7 measures a different object — a trajectory, not a matrix — and therefore extends the question space by a declared channel rather than by a family in Eq. (42). Let Φ_{dyn} map a per-item, per-cycle coverage log to

$$s_{dyn} = [\, f_{back},\ f_{multi},\ k_\infty,\ shape\left(\hat{p}(k)\right),\ r_{telescope}\,],$$

the observed backward-transition fraction, the multi-step fraction, the absorbing state, the fitted per-k flip-rate profile, and the residual of the telescoping identity of D3. The reference family M_{dyn} — *absorbing gradient flow on a bounded lattice* — is generated as $Binomial\left(9-k, \hat{p}(k)\right)$ ascent from the observed cycle-1 k-histogram, and the two matched nulls are a time-reversal null (relabel each transition's direction) and a cycle-order permutation null. Measured on the shipped log, $f_{back} = 0/598$, $k_\infty = 9$, and $r_{telescope} = 0.0$; the two matched nulls above are specified here and not executed, so no excluded/not-excluded verdict is issued on this channel. Any equilibrium family is excluded outright on the analytic argument of D2, not on a null. Source: tests/T3-results.json.

One boundary must be stated. The log covers 213 files over 6–7 cycles; the coverage matrix has 2286 rows at a different granularity. The dynamics channel therefore *corroborates* the append-only mechanism proposed for the row-mass bimodality of Section 6 on a subset at a coarser index, and does not confirm it row by row. The caveat in Section 8 stands.

Dynamics: the corpus's own construction

The scope section of the previous draft named a detailed-balance test on the atom edit log as the one clean way to separate potential from non-potential dynamics. That test is run here, on the shipped log (213 files, 6–7 cycles per corpus, 1391 dispatches, 1178 file-level cycle-to-cycle transitions; rsi-85 excluded as a known relabel of the 79). The state variable is the coverage count $k = 9 - |missing\left(i, c\right)| \in \{0, \ldots, 9\}$.

D1 — the log is strictly append-only, and this is a measurement, not bookkeeping. Of 1178 transitions, 580 stay (all of them the absorbing $k = 9$), 423 advance by one, 175 (14.9%) advance by 2–4, and *none* regresses. The missing sets are nested in 1178/1178 transitions; crucially, 175 of the 598 advancing transitions remove primitives *other than* the one the dispatch nominated, so coverage is re-assessed against the file each cycle rather than carried forward as a union. These 175 are the same set as the 175 multi-step advances counted above (per corpus: refs 147, skills 16, docs 12), not an independent tally: no single-step transition is shown to remove a non-nominated primitive; the re-measurement argument rests on the multi-step transitions alone, which a carried-forward union cannot produce. Monotonicity is therefore an empirical property of the corpus and its editor, not an artifact of the log's data structure. It also refutes a claim made elsewhere in this lineage: dispatches are counterfactual best-flip measurements, one per file per cycle, and the applied edit adds up to four primitives at once. Every transition out of $k = 0$ is +4 (14/14) and out of $k = 1$ is +3 or +4 (27/27). "Append-only" must not be read as "one primitive at a time".

D2 — detailed balance is violated maximally; no equilibrium free energy exists. $T(k+1 \to k) = 0$ for every k, in every corpus: upward mass 598, downward mass 0. Reversibility requires $\pi(k)T\left(k, k+1\right) = \pi(k+1)T\left(k+1, k\right)$ with the left side positive for $k = 2, \ldots, 8$ and the right side identically zero, so *no* positive π makes the chain reversible; the unique stationary distribution is $\delta_{k=9}$. The Kolmogorov cycle criterion is vacuously satisfied — with no backward edges the transition graph is a DAG — and is therefore uninformative, which is itself worth recording as an instance of Section 6.5. The entropy-production estimator $\sum_k J(k) \log\,[T(k \to k+1)/T(k+1 \to k)]$ diverges; it grows as $-423 \log \varepsilon$ under regularization and no single ε should be quoted. The defensible finite statement is a bound: with zero backward events in 598 opportunities, the rule of three puts the backward rate below 0.0051 at 95%, giving entropy production $\gtrsim 2.2 \times 10^3$ nats per sweep of the log ($423 \times \ln(1/0.0051) \approx 2233$ nats, one sweep being the 423 single-step advancing events). Any Ginzburg–Landau reading requires an equilibrium ensemble; there is none.

D3 — there is nonetheless a potential, and it is an identity. Define $\Phi(k) = d_{pre}(k)$. On the shipped log d_{pre} takes exactly one value per k in all three corpora ($\Phi = 2.0000, 1.9758, 1.9080, 1.8091, 1.6933, 1.5727, 1.4552, 1.3454, 1.2451$), and $d_{post}(k) - \Phi(k+1) = 0$ exactly for $k = 0, \ldots, 7$ (max absolute residual 0.0). Hence $\delta(k) = \Phi(k) - \Phi(k+1)$ and per-file cumulative Δ telescopes. This is not a discovered conservation law: the ladder distance is a fixed function of the coverage *count*, so the gradient property follows from the definition, and the empirical content is only that the distance is *exchangeable* in the nine primitives — it depends on how many are missing, not which. Two consequences follow. First, cumulative Δ is bounded and must decay to zero: "emergence saturating" is arithmetic on a bounded lookup. Second, the refs cycle-2 Δ peak (+11.98), the program's most-cited dynamical event, decomposes as a policy term of −0.274 (*fewer* dispatches: $109 \to 106$) against a state term of +2.368: refs begins at mean $k = 3.01$ and cycle 1 pushes the population to 5.57, straight onto the maximum of the fixed ladder at $k = 4$–5 ($\delta = 0.1206, 0.1175$). The peak is predictable in advance from the initial k-histogram; skills79, which starts at mean $k = 7.22$ past the hump, never shows one.

D4 — "potential vs non-potential" was the wrong axis. D2 and D3 hold simultaneously. A detailed-balance test can reject equilibrium; it cannot decide whether dynamics is potential in the gradient-descent sense, and the dichotomy proposed in the previous draft's scope section conflated the two. The restatement this program should carry is therefore stronger and narrower than the one proposed: *deterministic, exchangeable descent of a bounded coverage potential to an absorbing fixpoint, with zero back-flux and no equilibrium ensemble.*

D5 — what the absorbing-chain fit does and does not buy. A per-k independent-flip model (jumps $\sim Binomial\left(9 - k, p(k)\right)$, 9 parameters, AIC 878.15) beats a global-p model ($p = 0.4732$, AIC 1202.10; $2\Delta \log L = 339.96$ on 8 df) and a two-parameter saturation model whose exponent collapses to zero and which degenerates onto global-p. The fitted $\hat{p}(k) = 0.444, 0.463, 0.450, 0.343, 0.271, 0.367, 0.479, 0.536, 1.000$ is U-shaped with its minimum at $k = 4$, exactly where the per-flip payoff δ is maximal — flips are least likely where they are worth most — though k and cycle index are entangled ($\hat{p}$ also rises with cycle) and within cycle 1 alone the U-shape survives only shallowly. Two limits are stated rather than smoothed over: the 9 per-k rates are fitted on the same jump histogram used to score the trajectories, with no held-out check, and $\hat{p}(8) = 1.000$ is a boundary estimate. And *no* independent-flip model reproduces the terminal fixpoint: every such model leaves a geometric tail of 1–5 open files at the last cycle where the data show exactly zero. That sharpness is partly a stopping rule — the loop was run *until* $\Delta = 0$ — so the last cycle is conditioned on termination and should not be scored as a free prediction.

Sources for this section: tests/T3-results.json, tests/dynamics-series.json, regenerated by tests/t3_dynamics.py.

Bringing the free energy back: the curve-compass ± atom

D2 is a statement about *this log*: in 598 advancing opportunities the corpus's editor never once stepped backwards, so no positive π makes the measured chain reversible and there is no equilibrium ensemble for a Landau expansion to be an expansion of. That is a fact about a corpus, not a theorem about the ladder. Nothing forbids *designing* a different dynamics on the same measured $\Phi(k)$ whose equilibrium ensemble exists by construction, and then asking what the free-energy language buys once it is legitimately available. This subsection does exactly that, and is scrupulous about which side of the wall each number lives on.

The mechanism. The state is a configuration $\sigma \in \{0,1\}^9$ of covered primitives, $k = |\sigma|$. The *quantized ± atom* proposes one primitive flip: with probability 1/2 it turns a currently-off primitive on ($k \to k + 1$, an atomic positive), otherwise it turns a currently-on primitive off ($k \to k - 1$, an atomic negative), the primitive being chosen uniformly within the selected direction. Every accepted move therefore changes k by exactly ±1 and the Hamming distance by exactly 1 — quantization is a property of the move set, not of a fitted parameter. Because D3 established that Φ depends only on the coverage *count* (the exchangeability identity), the induced process on k is a birth–death chain, and the induced stationary distribution is available in closed form:

$$(46)\quad \pi_T(k) = \frac{1}{Z(T)}\binom{9}{k}\exp\left(-\Phi(k)/T\right) = \frac{1}{Z(T)}\exp\left(-F_T(k)/T\right),$$

$$(47)\quad F_T(k) = \Phi(k) - T\log\binom{9}{k}.$$

Equation (47) is a free energy in the exact sense: an energy term $\Phi(k)$, read off the paper's own measured ladder, minus temperature times the configurational entropy $\log\binom{9}{k}$ of the coverage shell. The ladder used is $\Phi = \left(2.0000, 1.9758, 1.9080, 1.8091, 1.6933, 1.5727, 1.4552, 1.3454, 1.2451, 1.1547\right)$ for $k = 0, \ldots, 9$.

Detailed balance holds by construction. Writing q for the proposal and α for the acceptance probability, the chain uses the full Metropolis–Hastings ratio

$$(48)\quad \alpha(k \to k') = \min\left\{1, \frac{\binom{9}{k'}}{\binom{9}{k}}\exp\left(-[\Phi(k') - \Phi(k)]/T\right)\right\} = \min\left\{1, \exp\left(-[F_T(k') - F_T(k)]/T\right)\right\},$$

so $\pi_T(k)\,P(k \to k') = \pi_T(k')\,P(k' \to k)$ for every pair, at every $T > 0$, as an algebraic identity rather than an empirical finding. The chain is irreducible and aperiodic on $\{0, \ldots, 9\}$, hence π_T of Eq. (46) is its unique stationary distribution. The empirical tests below are therefore not tests of whether balance holds — it does — but of whether the shipped implementation realizes the chain it claims to.

Empirical verification. It does. Table 14 collects the run record. At $T = 1$, over 8 chains × 200,000 steps (800,000 post-burn-in samples), every one of the nine net fluxes $J(k) = c(k \to k+1) - c(k+1 \to k)$ sits inside its 3σ binomial band with $\max_k |z\left(J(k)\right)| = 0.010$; the occupancy matches the exact π_T of Eq. (46) at total variation 0.00227 ($\chi^2 = 34.20$ on 9 df). The contrast with the historical log on the identical axis is the point of the subsection: there the forward flux vector is $J_{hist} = \left(1, 4, 4, 8, 27, 56\right)$ with an identically zero backward vector, 0/598.

quantity	compass (± atom)	historical log
$\max_k \lvert z\left(J\left(k\right)\right)\rvert$, $T = 1$	0.010 (all 9 within 3σ)	undefined (no backward edge)
backward transitions	balanced at every $T > 0$	0/598
$TV\left(\text{empirical}, \pi_T\right)$	0.00227 (800,000 samples)	—
χ^2 occupancy (9 df)	34.20	—
split-$\hat{R}$ / ESS / τ_{int}	1.00008 / 42,906 / 11.19	—
stationary TV (selftest)	0.00296 signed, 0.00120 atom	—
verdict	detailed balance *not rejected*	maximally irreversible

Table 14. The designed ± atom against the measured log, on the same flux axis. Compass figures from tests/compass-run/balance.json and the –selftest block of results/curve-compass-results.json (8/8 assertions pass); historical figures from Section 7 D1–D2.

The sweep, and what it shows. Over 25 log-spaced temperatures in $\left[0.005, 2\right]$ (8 chains × 40,000 steps, seed 20260812), the argmax of π_T leaves the absorbing endpoint $k = 9$ at a crossover $T_\times = 0.041143$: below it the energy term of Eq. (47) wins and π_T peaks at full coverage, above it the entropy term wins and the peak moves into the interior, relaxing to the $Binomial\left(9, 1/2\right)$ mean 4.5. The two sharp thermodynamic diagnostics locate the same crossover independently: the susceptibility $|\chi(T)| = |d\langle k\rangle/dT|$ peaks at $T = 0.0368$ ($\chi = -35.43$), and the specific-heat analogue $C(T) = Var_{\pi_T}[\Phi]/T^2$ peaks at $T = 0.038304$ ($C = 3.4565$), within 7% of $T_\times$ (the exact gap is 6.90%). Monte Carlo tracks the analytic π_T across the whole grid: $\langle k\rangle_{MC}$ agrees with $\langle k\rangle_{exact}$ within 3 MC standard errors at every one of the 25 temperatures (worst case 2.13 MCse at $T = 0.128$), with a maximum total variation of 0.0062, and split-$\hat{R} \le 1.001$ throughout.

One sweep diagnostic is flagged rather than sold. $Var_{\pi_T}[k]$ does attain an interior maximum, at $T = 1.787431$ with value 2.254301 — but the $T \to \infty$ binomial asymptote is $9/4 = 2.25$, so the peak stands only 0.19% above the floor it decays to. It is real (the ladder's increments $\delta(k)$ are largest mid-lattice, which broadens π_T slightly past binomial) and it is reported, but it is a 0.19% effect and no weight is placed on it; $C(T)$ and $|\chi(T)|$ are the diagnostics that actually resolve the crossover. A paper that spent Section 3.4 dismantling a fluctuation peak it could not reproduce should not now cash one at the third decimal place.

The $T \to 0$ limit recovers the historical log. Since Φ is strictly decreasing on $k = 0, \ldots, 9$, every coverage-increasing move is downhill in Φ, so as $T \to 0$ its Metropolis factor tends to 1 while every uphill factor $\exp\left(-\delta(k)/T\right)$ tends to 0 : the compass degenerates to monotone ascent absorbing at $k = 9$ — which is precisely the measured log. This is verified numerically rather than asserted. At $T = 0.01$ the chain puts mass 0.99903 on $k = 9$ with a backward rate of 4.79×10^{-4} per step; at $T = 0.001$ there are *zero* backward transitions and mass 1.00000 at $k = 9$, matching the historical $k_\infty = 9$ and $f_{back} = 0/598$. In the opposite limit the chain relaxes to pure entropy: TV against $Binomial\left(9, 1/2\right)$ is 0.00146 with $\langle k\rangle = 4.4999$ against 4.5. The historical dynamics is thus recovered as the greedy zero-temperature corner of a one-parameter family whose interior is reversible — and note what does *not* vanish on the way there: what goes to zero as $T \to 0$ is the backward *rate*, not the balance. The chain is reversible at every $T > 0$.

Two design notes, stated because they are load-bearing. (a) The ±-direction proposal is *asymmetric* on configuration space — from a state with k ones there are $9 - k$ ways up and k ways down, each drawn with probability 1/2 — so the acceptance must carry the full Metropolis–Hastings correction, the Hastings factor $\binom{9}{k'}/\binom{9}{k}$ of Eq. (48). This is not a refinement. A bare $\min\{1, \exp(-\Delta\Phi/T)\}$ would be Metropolis on Φ alone, with no entropy term at all; its stationary distribution would pin at $k = 9$ for every T and the crossover of Eq. (47) would not exist. The entropy in the free energy *is* the Hastings factor. (b) The value $\Phi(9) = 1.1547$ is the ladder continuation d_{post} at $k = 8$, not the 0.0 recorded at $k = 9$ in the log, which is a bookkeeping sentinel for "nothing left to fix". Taking the sentinel literally would plant a spurious well of depth 1.245 at the top of the ladder, swamp the entropy term, and destroy the crossover. Both choices are disclosed here because both change the answer.

What this does and does not overturn. It does not reopen the Ginzburg–Landau reading of the corpus, and the exclusion of Section 9 stands unmodified. The historical corpus still admits no equilibrium ensemble: D2 is untouched, 0/598 is untouched, and no temperature was measured anywhere in the log. What the compass demonstrates is narrower and, in the idiom of Section 6.5, more disciplined: the free-energy language applies *exactly* — not approximately, not by analogy — to a *designed* reversible dynamics on the same ladder, and the crossover $T_\times$ is a property of that design. T is a knob the designer sets, not an observable of the corpus; there is no estimator of T from the log, and none is offered. The correct reading of $T_\times = 0.041143$ is "the temperature at which this designed chain would stop preferring full coverage", not "the corpus's critical temperature". Stated plainly: the compass relocates the free-energy language from a property of the corpus, where this paper shows it does not hold, to a property of designable dynamics, where it holds by construction.

Artifacts for this subsection: the skill skills/curve-compass-skill/ (SKILL.md, scripts/curve_compass.py, references/phi-ladder.md, references/worked-runs.md), the consolidated record results/curve-compass-results.json, and the raw run logs in tests/compass-run/. Re-check with python3.12 skills/curve-compass-skill/scripts/curve_compass.py –selftest (8/8, GREEN, exit 0) from the bundle root; the sweep and balance blocks are reproduced by the sweep and balance subcommands at the seed and step counts recorded above. Proof pack: P14.

Scope and honest caveats

The retraction is part of the record. Section 3.3 withdraws 0.8005 / $z = -45.8$ after two independent replications. The correct posture toward the numbers replacing it is identical: 0.709180 ± 0.001183 is reproducible from the shipped scripts and seed, and should be re-run rather than cited.

z is over-dispersed. The calibration block measures $sd[z(\Delta V_2)] = 1.320$ at $s = 0$ where a perfectly calibrated z would give 1.000, with a one-sided FPR of 8.3% at the nominal 5% threshold. Every z here should be read as inflated by roughly a third in units of sd. This does not touch conclusions at $|z| > 12$; it does mean that z between 2 and 4 — the +2.12 for skills, the +3.41 for β_1 against the matrix null, the +2.5 at $N = 79$ — must not be treated as decisive.

Hodge magnitudes are construction-dependent. Only the directional statements of Table 8 are corpus properties. ρ_h spans $32\times$ and β_1 spans $2.2\times$ across four defensible constructions of the same matrix, and the sign of the ρ_c verdict flips. Reporting $\beta_1 = 6756$ as a corpus scalar would be an artifact; reporting "β_1 sits 16–313σ below its degree-matched null in every construction tried" is not. Separately, rank B_2 is computed by GF(2) elimination on the large graphs; it agreed with exact real rank wherever a dense SVD was feasible (synthetic $N = 300$: $991 = 991$; coverage-band graph: $2829 = 2829$), but 2-torsion in the clique complex would break that agreement and could not be excluded on the k-NN and threshold graphs. Graph-construction artifacts that mimic transitions — the Kahle density window — remain the first-class risk of this channel, which is why every Hodge number here carries a degree-matched null.

The sensitivity floor is $s \approx 1$ log-odds. At $N = 512$, $d = 9$, amplitude $s = 0.5$ gives $\Delta V_2 z = +0.04$ and power 0.00; $s = 1.0$ gives +3.43 and power 1.00. A non-detection at these N is a statement about power, not about the world, and must be reported as one. The generator's own threshold at $N = 200$ is $s = 0.75$, and thresholds do not transfer across N, d, base rate, or mode count.

What the Hodge nulls do not settle. Against the matrix null every Hodge quantity is within ~2.5σ, so the honest summary is that the coverage matrix's marginals plus the chosen graph jointly determine the entire Hodge profile. Anti-cyclicity is a statement about which items are near which items, not about latent capability organization beyond marginals. The Ginzburg–Landau bridge should be retired on this evidence: there is no free energy, no equilibrium dynamics, no ξ, no λ; the Hodge channel does not repair κ, and ρ_c is a disorder measure, not a temperature. Section 7 closes that door from the other side as well: the corpus's own construction dynamics has zero back-flux in 598 opportunities and admits no reversible π, so there is no equilibrium ensemble for a Landau free energy to be an expansion of.

Items closed. The previous draft named five untested items. All five are closed here, four of them negatively.

item	disposition
P-H3 clause (ii) — quantized azimuthal winding	**void**, and empirically not quantized: 2π is not a unit in Jaccard-normalized dominance flow, and all 14,158 cycles give $round\left(\oint h/2\pi\right) = 0$ with $\max|\oint h| = 0.1010$ (1.6% of 2π) — §4, §6.5
P-H2 — the 24-D curl clause	**not supported**: $\rho_s = +0.17$ and +0.10, $p \geq 0.69$, $n = 8$; low power (excludes only $\rho_s \gtrsim 0.7$) and a weak venue — §4
GL P4 / P5 / P7	**falsified** / **falsified** / **not supported**: no fluctuation peak once normalized by the null's own variance; a smooth power law with no divergence (and $1/N$ rejected too); no evidence for a critical exponent — §3.4, Table 6
ordering-permutation null (Remark 1)	**executed**, 500 shuffles: the azimuthal channel is dead and the direction is reversed — below
dynamics on the edit log	**closed**: absorbing gradient flow, zero regressions in 1178 re-measured transitions, no equilibrium ensemble — §7

Table 15. The five items the previous draft listed as untested, and where each now stands. None closes in the program's favour.

The ordering-permutation null, executed. Remark 1 required it before any azimuthal claim, and it is run here on both real corpora: 500 shuffles of which row receives which Fibonacci lattice point, with the matrix and the lattice untouched, so only the index map is randomized. The Y_3^3 probe is dead on both the 2286×9 master matrix and the real 269×9 GWTC matrix, under both weightings tried ($|z| \leq 0.19$, $p \geq 0.85$; on the master matrix under PC1 weighting specifically, the real value sits within a twentieth of a null sd of the shuffled mean, $|z| = 0.006$). The stronger result is the sign. The Y_3^3 Parseval share and the $m = 3$ share (both signs) sit *below* their nulls on both corpora (the $m = 3$ share at $z = -1.74$, -1.74, -1.56, -1.54 across the four weightings, i.e. a range of -1.54 to

-1.74; the Y_3^3 share at -1.06 to -1.29), and the mechanism is known: ordering by PC1 puts nearby rows at nearby lattice indices, which loads the low-m harmonics and starves $m = 3$, while a shuffled order scatters power into high m. The shuffled corpora are the ones with three-fold structure. This is not a non-detection; the effect runs the other way, and that is the definitive burial of the three-fold narrative in this lineage. One nominal hit survives to be reported and dismissed: GWTC P_3 under SNR weighting, $z = +2.35$, $p = 0.038$ — one of the 10 tests reported in this block, Bonferroni $p_{adj} = 0.38$, and it flips to $z = -0.96$ the moment the weight is changed to PC1 score on the same data, at $N = 269$ where power is low in any case. Source: tests/T1-results.json.

Provenance of the coverage labels. The zero-regression result of Section 7 is a re-measurement, not bookkeeping — the missing sets are re-assessed against each file every cycle, and 175 of the 598 advancing transitions remove primitives other than the one the dispatch nominated. What cannot be verified from the shipped log is *how* that re-assessment was made: the per-cycle coverage assessment is inherited from the improvement loop and its blinding is not documented, so if the same judge re-scored each file with the previous cycle's label in view, monotonicity could be partly manufactured by anchoring. The zero-regression result is conditional on that assessment being independent across cycles.

What is inherited and not re-verified. The predecessor's atom-level results (the empirical non-negativity of Δ over $1391 + 124$ dispatches with zero violations; the k-quantized Δ ladder peaking at $k = 5$) are not re-examined here, nor is the matched-parameter ablation, which remains two single-seed point estimates without error bars. Nothing in this paper contradicts them; nothing strengthens them either. Finally, the corpus's own row-mass bimodality is partly manufactured by the improvement loop that produced it — an append-only policy saturating skills, docs and refs while self stayed sparse — which is a plausible mechanism for the mixture placement of Section 6 but has not been traced row by row and is flagged as unconfirmed. Section 7 measures that append-only policy directly, but on 213 files over 6–7 cycles against the coverage matrix's 2286 rows at a different granularity: it *corroborates* the mechanism on a subset at a coarser index and does not confirm it row by row, and the flag stands.

Fold-slope false positives are empirical, not theoretical. The t_{fold} of Eq. (38) is *not* referred to a t distribution. Its inputs are the same over-dispersed per-cell z ($sd = 1.320$ at $s = 0$), and four of them per trial are neither independent nor identically scaled; the only defensible reference is the empirical null ladder, whose 95% quantile is 0.0997 at this $\left(N, d, T\right)$. Quoting $t_{fold} = 45.19$ against a Gaussian tail would be exactly the error this paper indicts elsewhere. The quantile is design-specific and does not transfer across N, d, K or the ratio ρ.

The real-GWTC z is inflated by parameter redundancy. $V_2 = 0.835$ at $z = +67.0$ (Section 6.4) is measured on nine fields of which chirp, total and final mass are functions of $\left(m_1, m_2\right)$, and redshift a function of luminosity distance. The correlation matrix is numerically rank-deficient ($\lambda_7, \lambda_8, \lambda_9 = 0.0010$, 0.0004, ≈ 0), and a column-shuffle null destroys precisely the dependence those identities guarantee. The number is correct and its interpretation is narrow: it certifies redundancy, not astrophysical structure. No claim about black-hole populations is made or supported here.

Azimuthal claims remain unsupported on real data. The Y_3^3 Parseval share on the real catalogue is 0.00334 against a null 0.00832 ± 0.00427, $z = -1.17$. That is a non-detection, and with 269 events it is also a low-power one — it excludes nothing about the sky, and it supports nothing either. The two items that were still open when that sentence was first written are now closed and point the same way: the ordering-permutation null of Remark 1 has been executed and leaves the Y_3^3 probe at $|z| \le 0.19$ with the $m = 3$ energy shares *below* their nulls for a stated reason, and P-H3 clause (ii) is void as posed and empirically not quantized. Every three-fold-symmetry claim in this program's lineage is therefore unsupported, and on the azimuthal-share coordinate the measured effect runs opposite to the claim.

Conclusion

The program's headline observable is, to 98%, a function of its own marginals. Its celebrated transition is a change of basis on identical rows (Proposition 1). Its gate is a rank test that a corpus of pure noise passes at exactly 1.0000 in two of the eleven bases of Table 1 (Proposition 2), and the "effective rank" reported beside it is $2/V_2$, a restatement of the gate rather than a check on it. Its critical point disappears when the null's own finite-size behaviour is subtracted. Its Hodge reading is inverted: the corpus is anti-cyclic, with β_1 some 313σ below a degree-matched null, and all three pre-registered predictions of that channel are falsified. And its most-cited intermediate correction — the $z = -45.8$ reversal — was itself wrong and is retracted here on the strength of two independently validated uniform samplers.

What survives, measured rather than asserted, is a residual of $\Delta V_2 = +0.0144$ at $z = +12.3$: small, real, and placed. And what replaces the sequence of candidate X's is the map of Eq. (1) together with the instrument that calibrates it — a generator whose planted amplitude is known, on which $\Delta V_2 z$ has power monotone in signal above $s = 0.5$, is quiet (though over-dispersed) at zero signal, and powered to 1.00 at one log-odds by $N = 512$, while the statistics the program previously reported (V_2, the gate, the sphere-fit R^2) are flat, degenerate, or anti-ordered over the same sweep. yubiOS is then not "partially compatible with Ginzburg–Landau". It is: M_0 excluded at $z = +12.3$, mean-field GL excluded on the shape of $\Delta V_2(N)$, nearest family a two-population mixture, no excess cyclicity, off-map families named as not-tested rather than as partial. The next corpus can be placed on the same map by the same script, and the map's own round trip — planted excludes the null at $z = 27.9$, recovers its own family at $\max|z| = 0.99$ — is what makes that placement mean anything.

Is this corpus X? For every X this program named, no. Mean-field Ginzburg–Landau is excluded on the shape of $\Delta V_2(N)$ and now on three further predictions run rather than asserted — no fluctuation peak beyond the subsampling baseline, no susceptibility divergence, no evidence of critical slowing-down — and it is excluded in principle as well: the corpus's own construction dynamics admits no equilibrium ensemble, so there is no free energy for a Landau expansion to be an expansion of. It is not a vortex system: the corpus is anti-cyclic, with β_1 some 313σ below a degree-matched null and zero harmonic support on the defect rows. It is not three-fold symmetric: under an ordering-permutation null the Y_3^3 probe is dead on both the master matrix and the real GWTC catalogue ($|z| \le 0.19$), and the $m = 3$ energy shares sit *below* their nulls, because ordering by PC1 makes the fit smoother in azimuth — the shuffled corpora are the ones with three-fold structure. What the corpus is, positively and measurably, is two things. As a matrix it is a two-population coverage mixture: 98% of the headline observable is fixed by its marginals, and the surviving corpus-specific residual is $\Delta V_2 = +0.0144$ at $z = +12.3$, small, real, and nearest a latent two-class mixture. As a process it is an absorbing gradient flow: 1178 independently re-measured transitions with zero regressions, coarse-grained (14.9% adding two to four primitives at once) and front-loaded, descending a bounded exchangeable potential in the coverage count to a single absorbing fixpoint at full coverage, maximally irreversible and never returning. The second fact explains the first — an append-only policy that saturates one population while leaving another sparse manufactures exactly the row-mass bimodality that places the matrix off its own null — and it does so without any critical point, order parameter, or phase.

One coda, because it sharpens the exclusion rather than softening it. Put a *designed* reversible dynamics on the same measured ladder — the quantized ± atom of Section 7.1, one primitive flip per move with Metropolis–Hastings acceptance — and the free-energy language returns exactly, with $F_T(k) = \Phi(k) - T \log\binom{9}{k}$, detailed balance not rejected at $\max_k |z\left(J\right)| = 0.010$, an energy–entropy crossover at $T_\times = 0.041143$, and a $T \to 0$ limit that reproduces the historical absorbing log. That is not a reprieve for the corpus. It is the cleanest possible statement of what was wrong with the original reading: the Ginzburg–Landau vocabulary was never unavailable in principle, only unavailable *here* — it attaches to dynamics one designs, where T is a knob, not to the corpus one measured, where no temperature exists to estimate.

Appendix A: Reproduction manifest

All paths are relative to the evidence bundle root. Master seed 20260812; Python 3.12.12, numpy 2.5.1, scipy 1.18.0. Null convention: fixed-margin curveball ($10N$ trades unless stated; the headline V_2 null uses 200 samples × 228,600 trades) for binary matrices, independent column permutation for continuous ones.

Scripts (scripts/).

common.py (shared loaders, V_2, the two-share proxy $\hat{r} = 2/V_2$ of Eq. (15), spectra); null_models.py (curveball, swap-chain, column permutation, iid, row-mass-only; sampler uniformity harness); hodge_decompose.py (Eqs. (21)–(30): boundary matrices, LSMR solves, $\rho_\bullet$, GF(2) Betti numbers, orthogonality assertions, hexagon control); run_real_corpora.py (per-corpus V_2/Hodge against both nulls); audit_curveball_discrepancy.py (the retraction audit of Remark 2: fibre enumeration, χ^2 uniformity, mixing sweep, cross-checks against the other three nulls); generate_samples.py (planted / mixture / null corpora, Eqs. (9), (34)–(36)); ladder_correlations.py (Table 1); timeseries_build.py (consolidated series); p4_variance_ratio.py (Table 6: the P4 sweep normalized by the curveball null's own variance at each N, with the $(1/N - 1/2286)$ baseline printed alongside; ~35 s at the defaults); fold_ladder_experiment.py (Empirical Result 2 and Table 10: constant-ratio ladder, paired/unpaired designs, fold slope of Eq. (38), null ladder; imports the instrument by relative path; ~2–4 min at the defaults); real_gwtc_pipeline.py (Section 6.4 and Table 12: GWOSC ingest with –fetch, the 269×9 build, V_2, the column-shuffle null, the sphere fit and the Parseval shares of Eq. (40); seconds).

Results (results/).

A-v2-nulls.json (Table 3); A2-curveball-audit.json (Table 2, mixing sweep, retraction evidence); validation.json and validation-curveball-sampler.json (sampler uniformity: $\chi^2 = 118.096$, $p = 0.506$; $\chi^2 = 123.758$, $p = 0.364$; Y_3^3 constants); B-hodge-yubios.json (Eq. (33), Table 7); C-predictions.json (P-H1, P-H2, P-H3); D-construction-sensitivity.json (Table 8); E-harmonic-support.json (harmonic-energy concentration); E1-hodge-nulls-report.md (the Hodge report in full, including failures); F-hodge-cycle18-knn8.json ($N = 1391$ replication); ladder_VD.json (Table 1); finite_size.json, finite_size_colperm.json, scaling_VN.json (Table 4, Eq. (18)); signal_recovery.json (Table 9 and the false-positive block); gwtc.json (GWTC-like corpora and nulls); manifest.json (per-corpus N, column sums, zero-row counts, source paths); fold-ladder-experiment.json (Table 10); real-gwtc-results.json (Table 12: V_2, null, sphere fit, Parseval shares, Y_3^3 null); curve-compass-results.json (Section 7.1 and Table 14: the Φ ladder with its provenance, the 8/8 selftest block, the crossover $T_\times = 0.041143$, the $\chi/C(T)/Var[k]$ peaks, and the full 25-point sweep grid); SUMMARY.md (ladder, recovery, N_c re-analysis, Y_3^3 constants).

Consolidated series (timeseries/new-series/).

INDEX.json (provenance, null convention, validation block); ladder-VD.json (77 rows, V_2 vs D); scaling-VN.json (31 rows, V_2 vs N, null-subtracted); signal-recovery.json (248 rows, planted-amplitude sweep plus FPR); gwtc-new.json (4 rows). Generated matrices are shipped in data/samples/ (family_i_latent_class.npz, family_ii_curved.npz, family_iii_gwtc.npz, MANIFEST.json); they are also regenerable from scripts/generate_samples.py in under a second.

Instrument (skills/curved-corpus-create/).

SKILL.md (conventions, gate failure mode, anti-patterns, red flags), scripts/create_corpus.py (subcommands generate, measure, calibrate, place; –selftest asserts seven blocks including the round trip of Eq. (45) and the K-invariance of Eq. (11)), references/ (worked calibration example). Dependencies: Python stdlib and numpy only, no network.

Compass instrument (skills/curve-compass-skill/).

SKILL.md (the designed-dynamics thesis, the by-construction detailed-balance argument, the two design notes of Section 7.1, and the anti-patterns that follow from them), scripts/curve_compass.py (subcommands simulate, sweep, balance, history; –selftest asserts eight blocks — stationarity against the exact π_T of Eq. (46), empirical detailed balance, split-$\hat{R}$/ESS/τ_{int} convergence, the $T \to 0$ absorption of Section 7.1, the $T \to \infty$ binomial limit, the fluctuation and heat-capacity peaks, $|\Delta k| = 1$ quantization, and bitwise determinism), references/phi-ladder.md (the Φ ladder and the $\Phi(9)$ sentinel question), references/worked-runs.md (worked sweep and balance runs). Dependencies: Python stdlib and numpy only, no network. Re-check: python3.12 skills/curve-compass-skill/scripts/curve_compass.py –selftest from the bundle root (GREEN, 8/8, exit 0); also run as CHECK 5 of proofs/VERIFY.sh.

Real inputs.

The 2286×9 master matrix is shipped at data/real/per_row_coverage_v3.json (column sums $\left[1640, 1304, 1208, 1296, 1098, 1320, 1776, 1373, 1204\right]$, 204 zero rows); it originated as cycle-20/per_row_coverage_v3.json and earlier drafts of this appendix cited it under that historical path. The 1391×9 replication input (cycle-18/per_row_coverage.json, 14 zero rows) is *not* shipped in the bundle; only the results derived from it are, in results/F-hodge-cycle18-knn8.json. The headline V_2 is computed over all 2286 rows, the 204 all-zero rows included; restricting to the 2082 non-zero rows gives 0.678655 instead of 0.723529, so the inclusion policy is part of the definition of the number. The real GWTC inputs are shipped at data/real/gwtc-real-gwosc.json (the GWOSC eventapi superset as fetched, 391 events) and data/real/real-gwtc-matrix.csv (the derived 269×9 matrix, header row naming the nine fields in order); scripts/real_gwtc_pipeline.py rebuilds the second from the first, or refetches the first with –fetch.

Compass run logs (tests/compass-run/).

selftest.txt (the 8/8 GREEN block quoted in Table 14); balance.txt, balance.json (the $T = 1$ flux table, $8 \times 200{,}000$ steps, $800{,}000$ samples: per-k forward/backward counts, 3σ binomial bands, $\max_k |z\left(J\right)| = 0.010$, $TV = 0.00227$, $\chi^2 = 34.20$); sweep.txt, sweep.json (25 log-spaced temperatures in $\left[0.005, 2\right]$, $8 \times 40{,}000$ steps, seed 20260812, with $T_\times$ and the three peak locations); history.txt, history.json (the historical-log re-measurement and the $T \to 0$ correspondence check); simulate_T005.txt (a single worked trajectory at $T = 0.05$).

Real inputs (continued).

Primitive order throughout: attestation, trust_chain, least_privilege, declarative_policy, continuous_adaptive, immutability, audit_evidence, cryptographic_identity, segmentation.

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